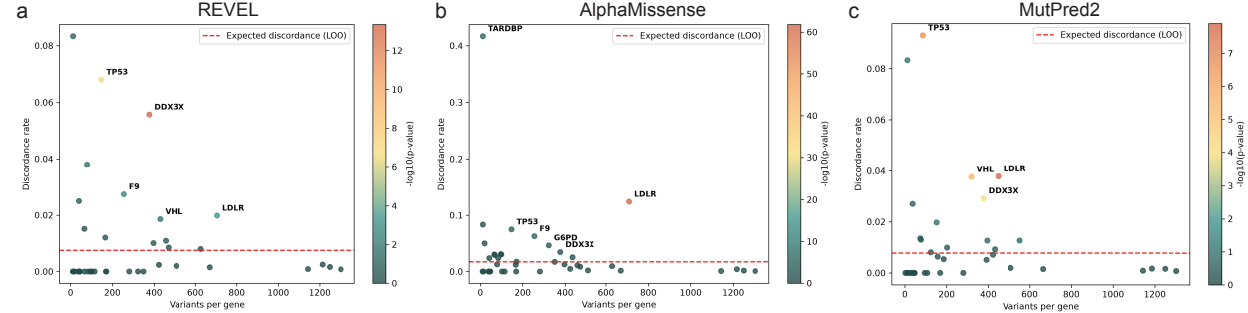
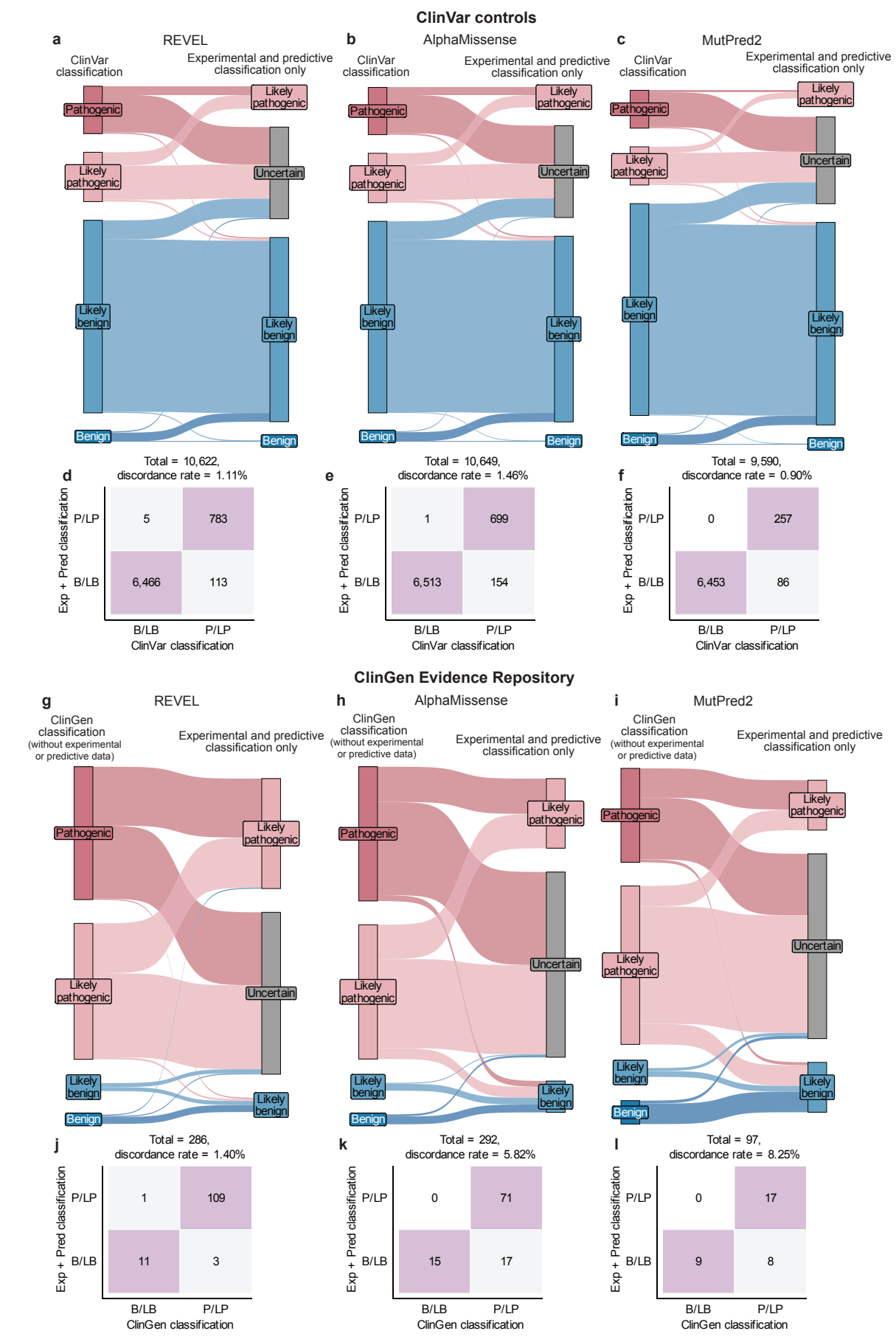
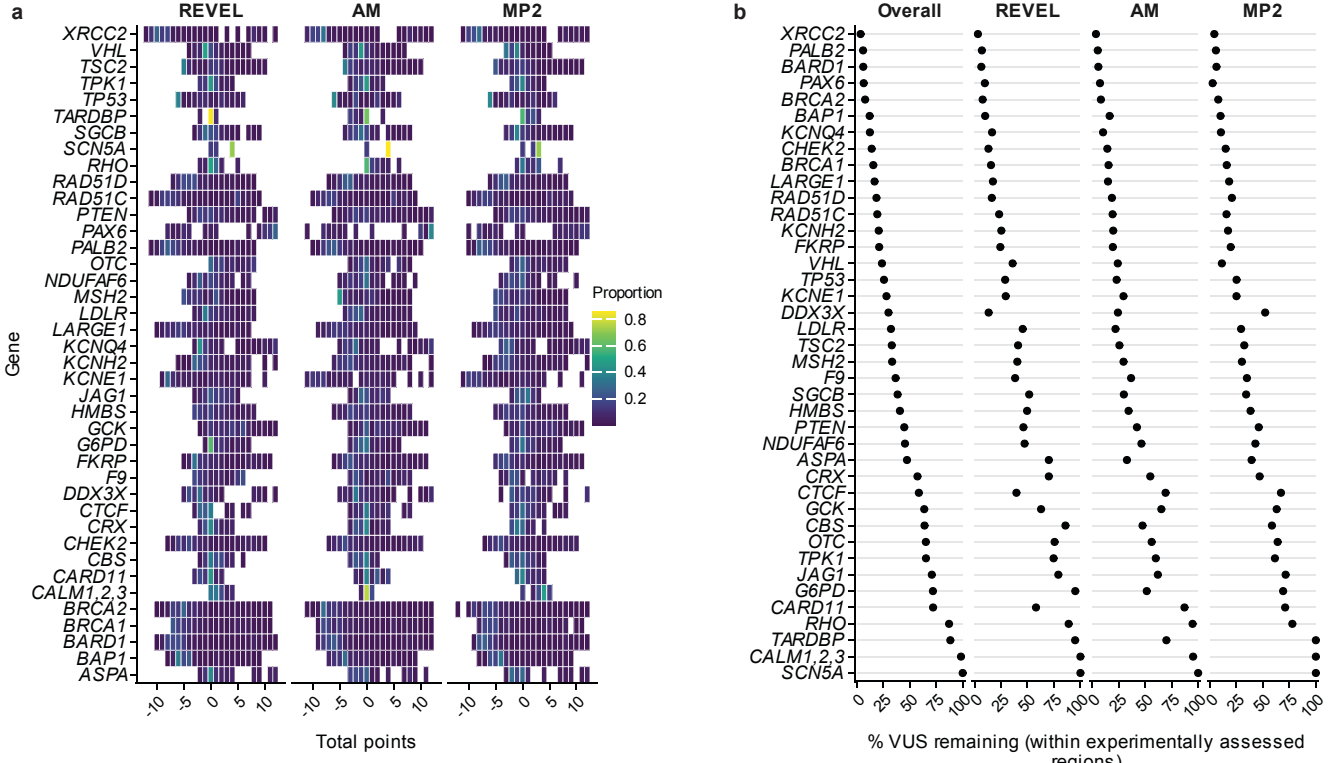
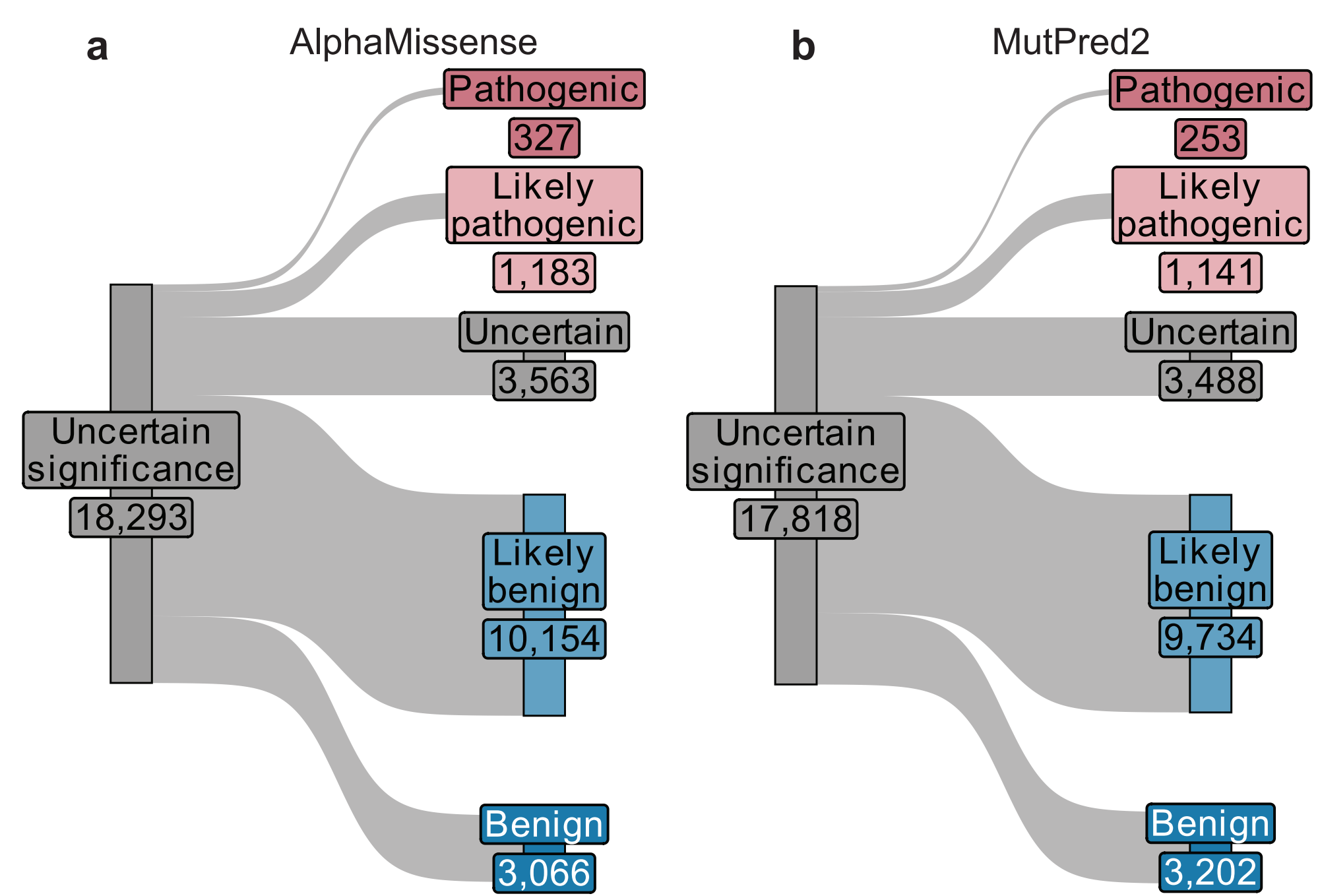


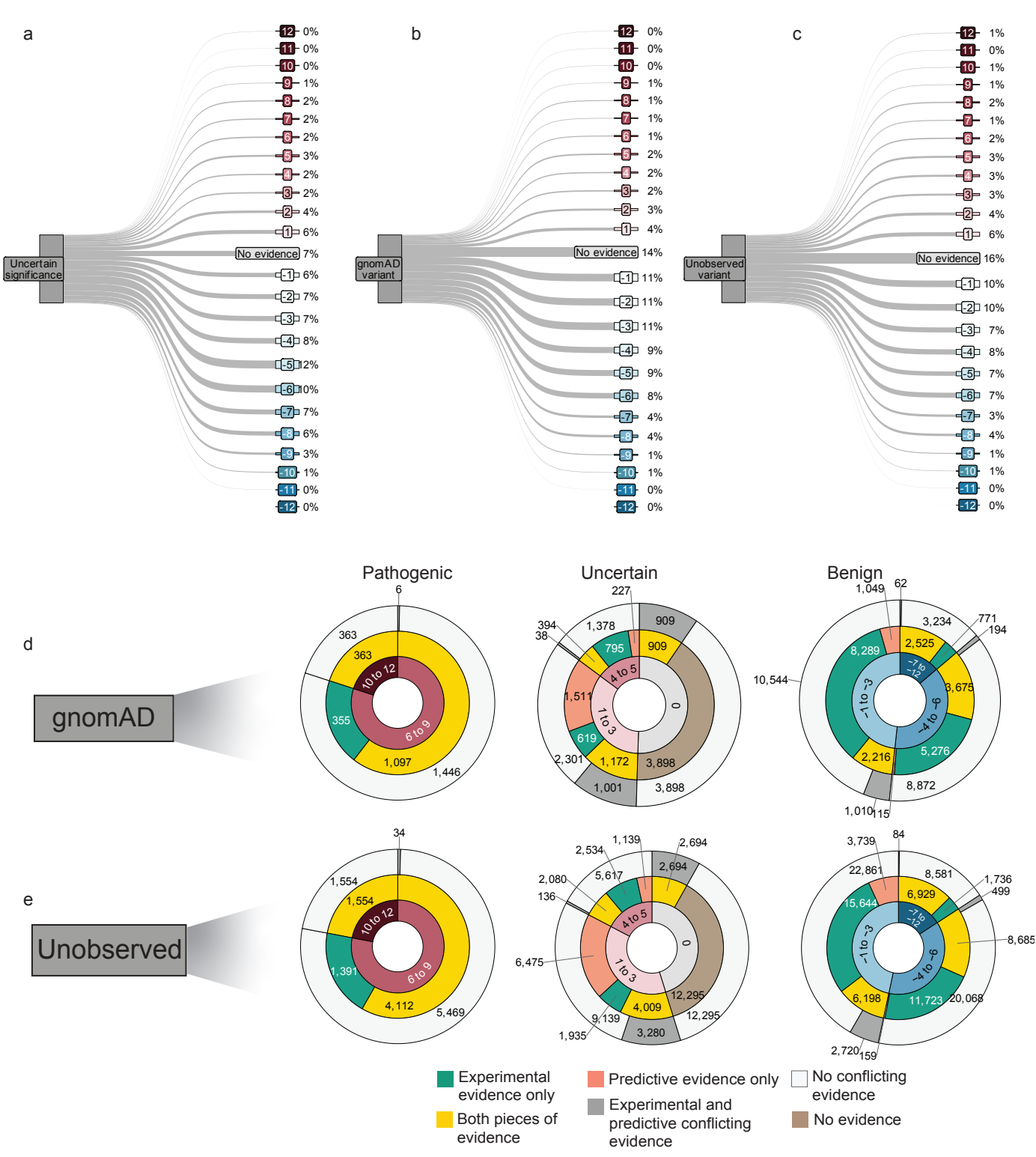
Extended Data Figure 4: Genes with the most discordance with ClinVar controls

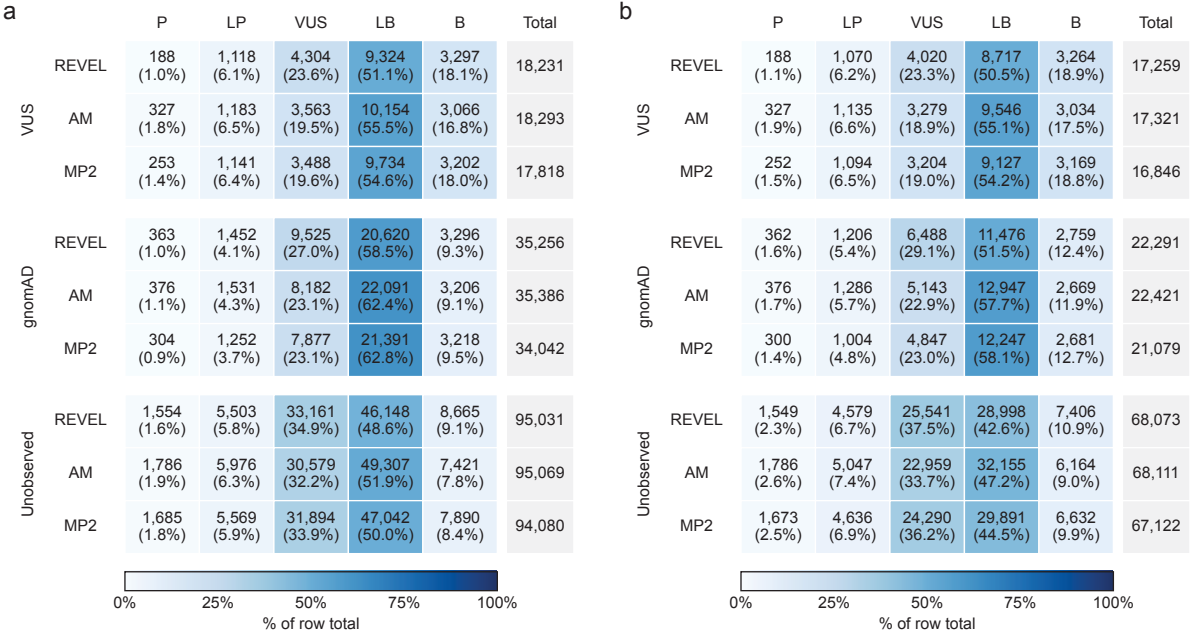
Gene-level enrichment of discordant variant classifications when using (a) REVEL, (b) AlphaMissense, or (c) MutPred2 as the predictor. Genes are plotted by variant count (x-axis) and discordance rate (y-axis; note the different scale in panel b) between ClinVar classifications and classifications using only experimental and predictive data (REVEL predictor). Discordance reflects pathogenic-benign classification flips when comparing ClinVar classifications and calculated classifications. The expected discordance rate (red dashed line) was estimated using a leave-one-gene-out background discordance. Colors indicate −log10(p-value) from a binomial test of excess discordance; significantly enriched genes (FDR q < 0.05, ≥10 variants) are labeled.


ClinVar controls
a
REVEL
ClinVar classification
Experimental and predictive classification only
Pathogenic
Likely pathogenic
Likely benign
Benign
Uncertain
b
AlphaMissense
c
MutPred2
d
Total = 10,622,
discordance rate = 1.11%
Exp + Pred classification
P/LP
B/LP
5
783
6,466
113
B/LB
P/LP
ClinVar classification
e
Total = 10,649,
discordance rate = 1.46%
1
699
6,513
154
f
Total = 9,590,
discordance rate = 0.90%
0
257
6,453
86
ClinGen Evidence Repository
g
REVEL
ClinGen classification (without experimental or predictive data)
h
AlphaMissense
i
MutPred2
j
Total = 286,
discordance rate = 1.40%
1
109
11
3
ClinGen classification
k
Total = 292,
discordance rate = 5.82%
0
71
15
17
l
Total = 97,
discordance rate = 8.25%
0
17
9
8

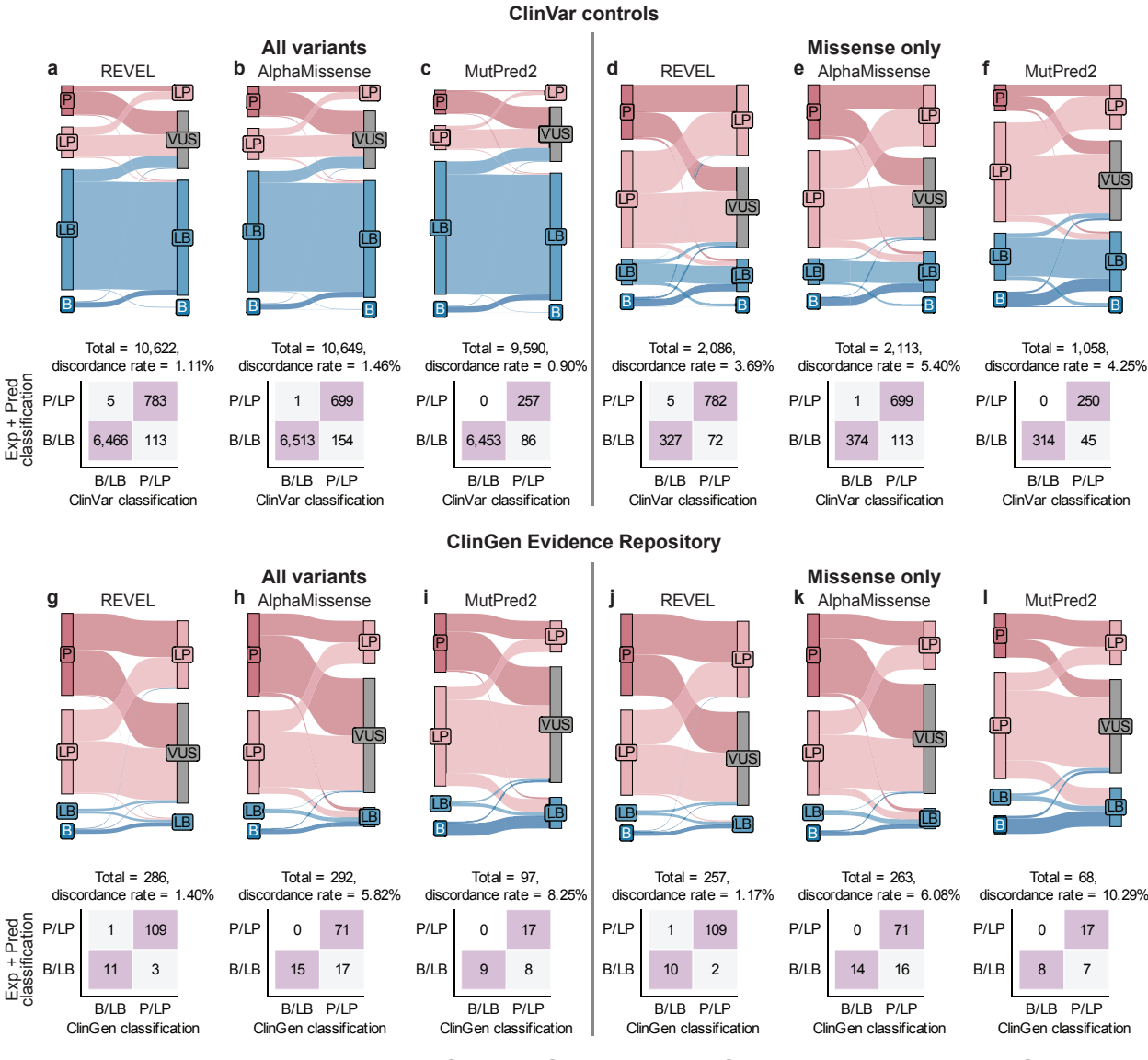

a
REVEL
AM
MP2
Gene
XRCC2
VHL
TSC2
TPK1
TP53
TARDBP
SGCB
SCN5A
RHO
RAD51D
RAD51C
PTEN
PAX6
PALB2
OTC
NDUFAF6
MSH2
LDLR
LARGE1
KCNQ4
KCNH2
KCNE1
JAG1
HMBS
GCK
G6PD
FKRP
F9
DDX3X
CTCF
CRX
CHEK2
CBS
CARD11
CALM1,2,3
BRCA2
BRCA1
BARD1
BAP1
ASPA
Proportion
0.8
0.6
0.4
0.2
-10
-5
0
5
10
Total points
b
Overall
REVEL
AM
MP2
XRCC2
PALB2
BARD1
PAX6
BRCA2
BAP1
KCNQ4
CHEK2
BRCA1
LARGE1
RAD51D
RAD51C
KCNH2
FKRP
VHL
TP53
KCNE1
DDX3X
LDLR
TSC2
MSH2
F9
SGCB
HMBS
PTEN
NDUFAF6
ASPA
CRX
CTCF
GCK
CBS
OTC
TPK1
JAG1
G6PD
CARD11
RHO
TARDBP
CALM1,2,3
SCN5A
0
25
50
75
100
% VUS remaining (within experimentally assessed regions)

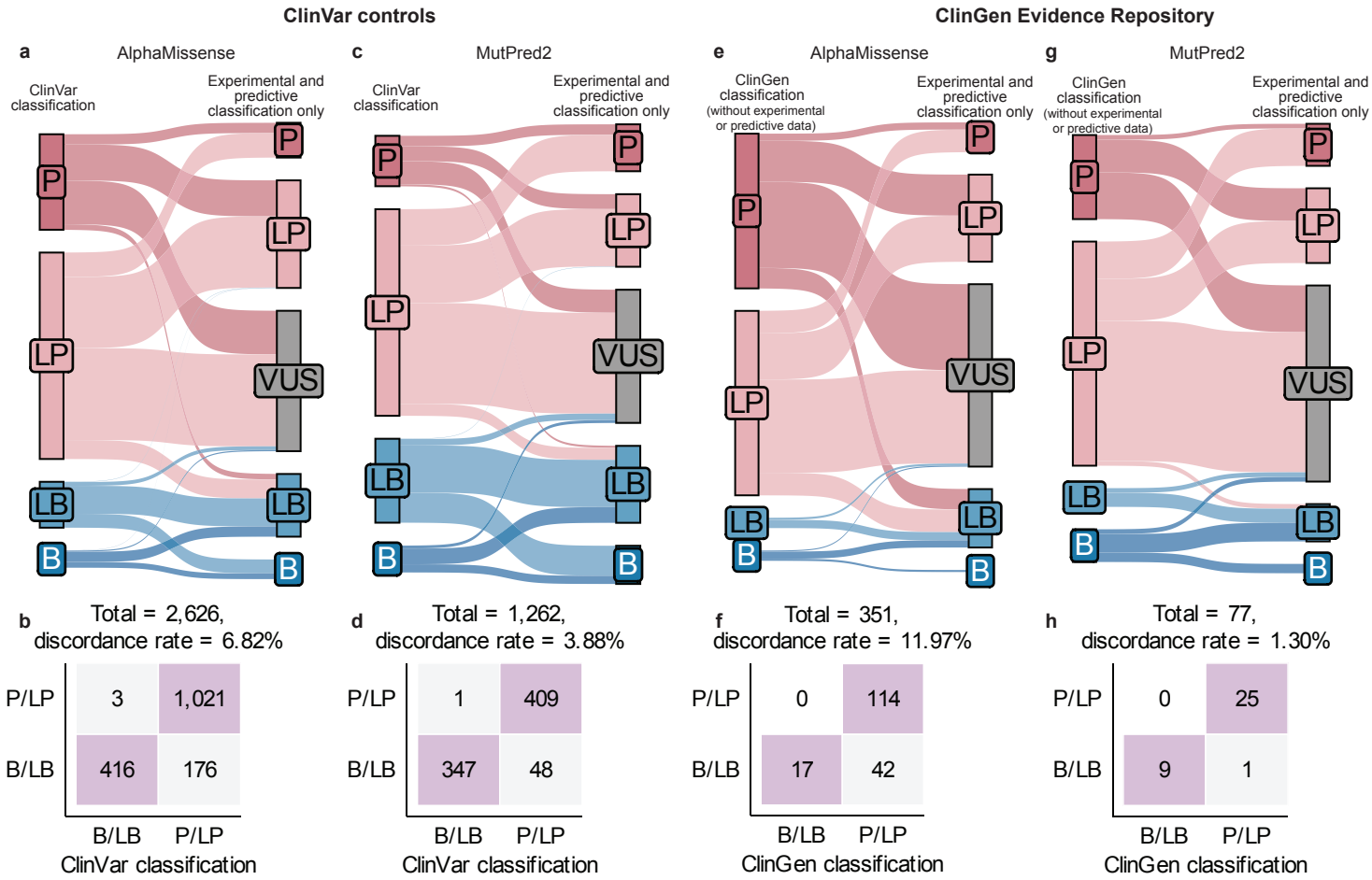

a
AlphaMissense
Pathogenic
327
Likely pathogenic
1,183
Uncertain
3,563
Uncertain significance
18,293
Likely benign
10,154
Benign
3,066
b
MutPred2
Pathogenic
253
Likely pathogenic
1,141
Uncertain
3,488
Uncertain significance
17,818
Likely benign
9,734
Benign
3,202

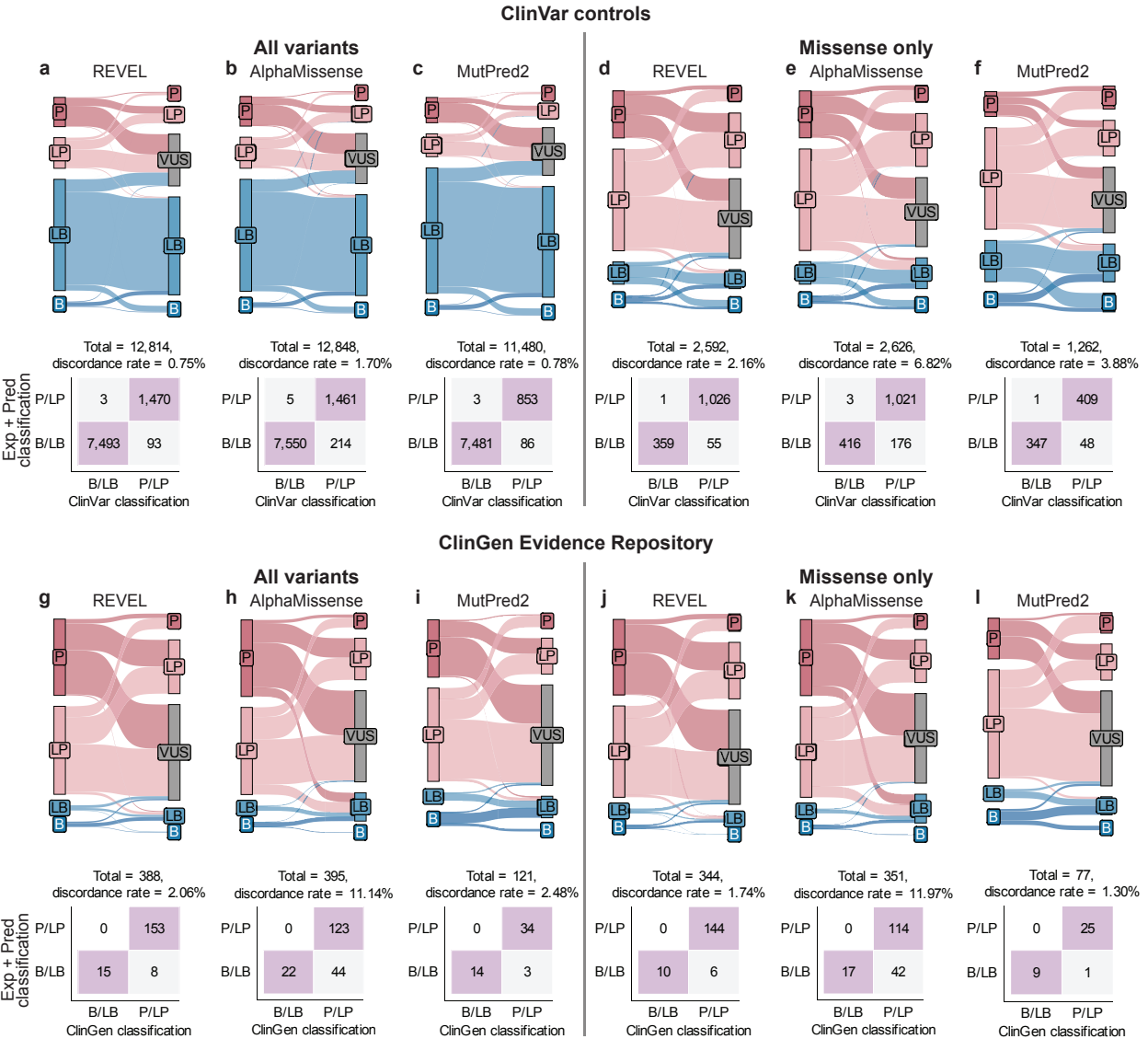

a
Uncertain significance
12 0%
11 0%
10 0%
9 1%
8 2%
7 2%
6 2%
5 3%
4 2%
3 2%
2 4%
1 6%
No evidence 7%
-1 6%
-2 7%
-3 7%
-4 8%
-5 12%
-6 10%
-7 7%
-8 6%
-9 3%
-10 1%
-11 0%
-12 0%
b
gnomAD variant
12 0%
11 0%
10 0%
9 1%
8 1%
7 1%
6 1%
5 2%
4 2%
3 2%
2 3%
1 4%
No evidence 14%
-1 11%
-2 11%
-3 11%
-4 9%
-5 9%
-6 8%
-7 4%
-8 4%
-9 1%
-10 1%
-11 0%
-12 0%
c
Unobserved variant
12 1%
11 0%
10 1%
9 1%
8 2%
7 1%
6 2%
5 3%
4 3%
3 3%
2 4%
1 6%
No evidence 16%
-1 10%
-2 10%
-3 7%
-4 8%
-5 7%
-6 7%
-7 3%
-8 4%
-9 1%
-10 1%
-11 0%
-12 0%
d
gnomAD
Pathogenic
Uncertain
Benign
6
363
363
355
10 to 12
6 to 9
1,097
1,446
227
909
909
394
38
1,378
795
4 to 5
1,511
1 to 3
619
2,301
1,172
3,898
1,001
3,898
0
1,049
62
3,234
771
194
10,544
8,289
2,525
-10 to -12
3,675
-1 to -3
-6 to -9
-4 to -5
5,276
2,216
8,872
1,010
115
e
Unobserved
34
1,554
1,554
1,391
10 to 12
6 to 9
4,112
5,469
1,139
2,534
2,694
2,694
5,617
2,080
136
4 to 5
6,475
1 to 3
0
12,295
4,009
12,295
9,139
1,935
3,280
3,739
84
22,861
8,581
1,736
499
15,644
6,929
-10 to -12
8,685
-1 to -3
-6 to -9
-4 to -5
6,198
11,723
20,068
2,720
159
Experimental evidence only
Both pieces of evidence
Predictive evidence only
Experimental and predictive conflicting evidence
No conflicting evidence
No evidence



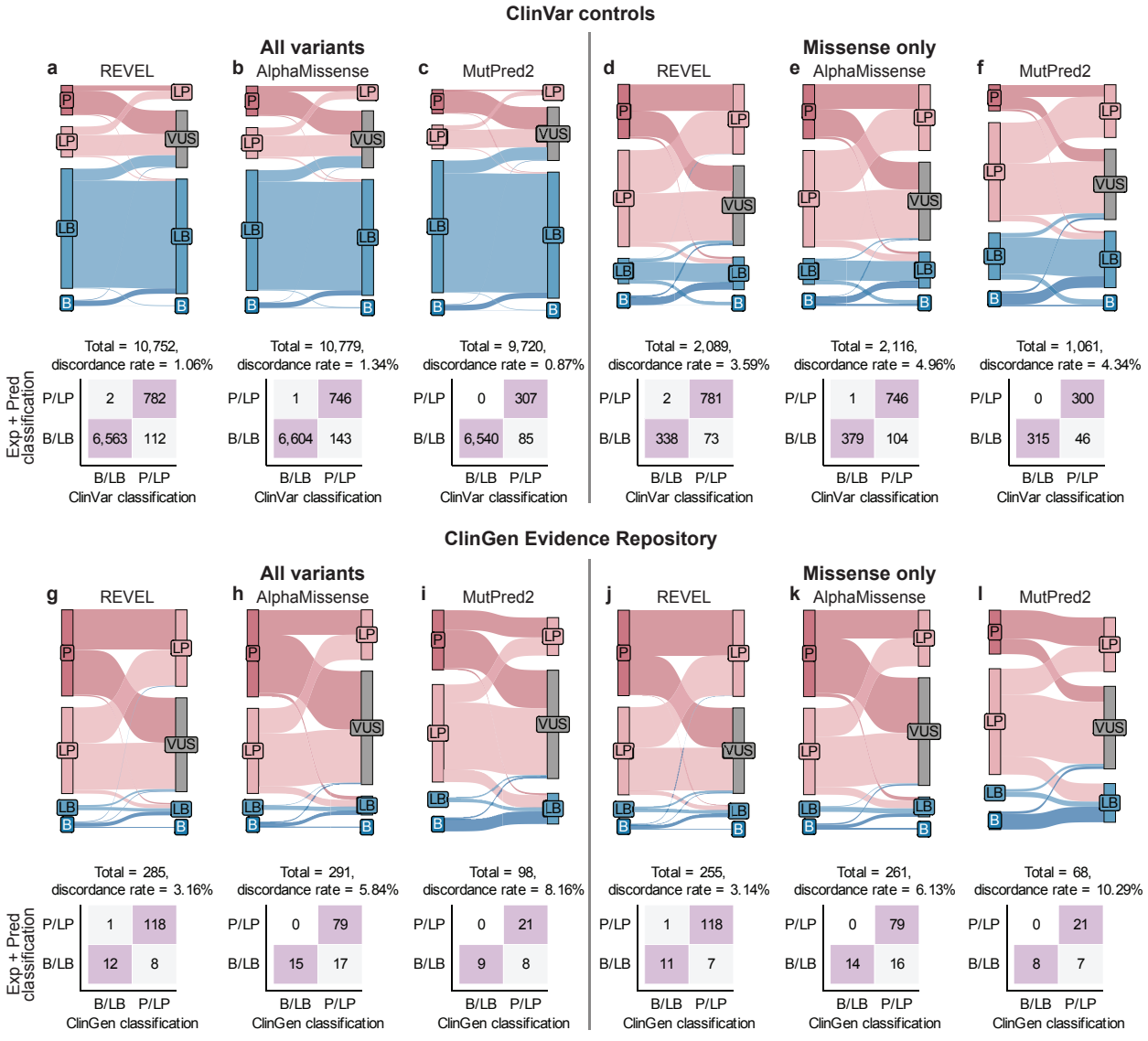
Extended Data Figure 2: Calibrating pathogenicity predictions

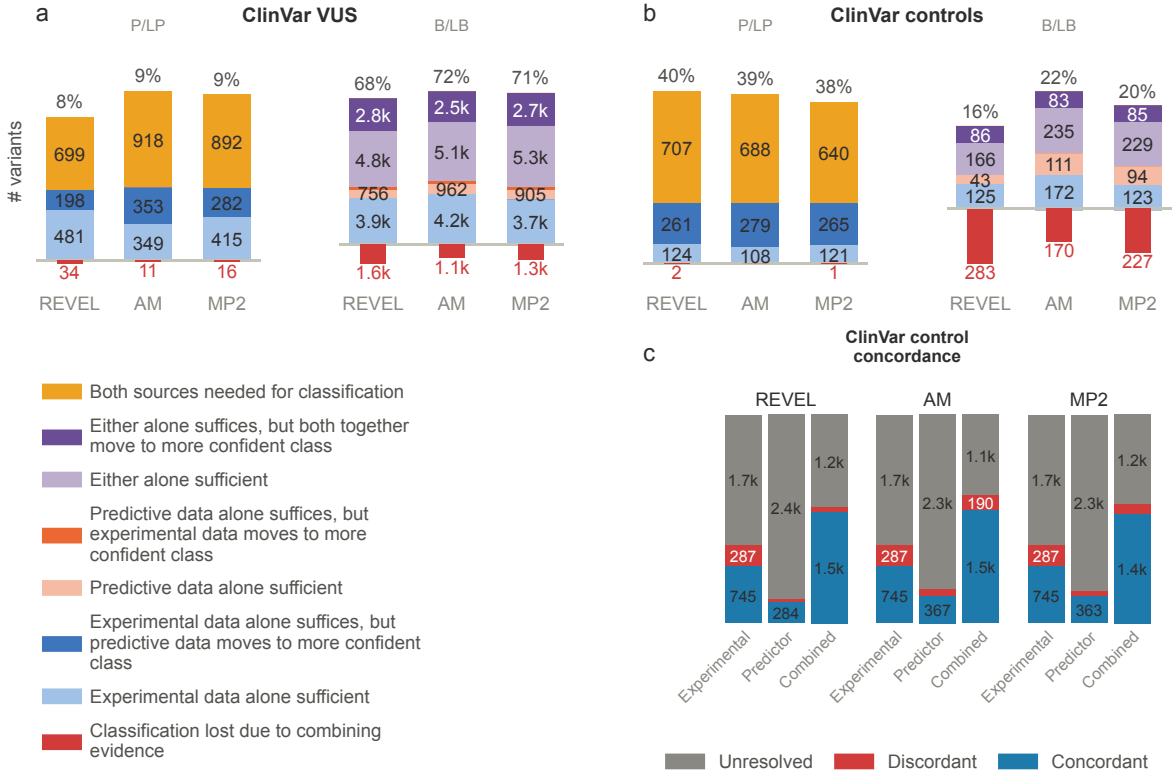

ClinVar controls
All variants
Missense only
a REVEL
b AlphaMissense
c MutPred2
d REVEL
e AlphaMissense
f MutPred2
P
LP
VUS
LB
B
Total = 10,622, discordance rate = 1.11%
Total = 10,649, discordance rate = 1.46%
Total = 9,590, discordance rate = 0.90%
Total = 2,086, discordance rate = 3.69%
Total = 2,113, discordance rate = 5.40%
Total = 1,058, discordance rate = 4.25%
Exp + Pred classification
P/LP
B/LB
5 783
6,466 113
1 699
6,513 154
0 257
6,453 86
5 782
327 72
1 699
374 113
0 250
314 45
B/LB P/LP
ClinVar classification
ClinGen Evidence Repository
All variants
Missense only
g REVEL
h AlphaMissense
i MutPred2
j REVEL
k AlphaMissense
l MutPred2
Total = 286, discordance rate = 1.40%
Total = 292, discordance rate = 5.82%
Total = 97, discordance rate = 8.25%
Total = 257, discordance rate = 1.17%
Total = 263, discordance rate = 6.08%
Total = 68, discordance rate = 10.29%
1 109
11 3
0 71
15 17
0 17
9 8
1 109
10 2
0 71
14 16
0 17
8 7
ClinGen classification


ClinVar controls
ClinGen Evidence Repository
a
AlphaMissense
ClinVar classification
Experimental and predictive classification only
P
LP
LB
B
VUS
c
MutPred2
ClinVar classification
Experimental and predictive classification only
e
AlphaMissense
ClinGen classification (without experimental or predictive data)
Experimental and predictive classification only
g
MutPred2
ClinGen classification (without experimental or predictive data)
Experimental and predictive classification only
b
Total = 2,626, discordance rate = 6.82%
P/LP
B/LB
3
1,021
416
176
B/LB
P/LP
ClinVar classification
d
Total = 1,262, discordance rate = 3.88%
P/LP
B/LB
1
409
347
48
B/LB
P/LP
ClinVar classification
f
Total = 351, discordance rate = 11.97%
P/LP
B/LB
0
114
17
42
B/LB
P/LP
ClinGen classification
h
Total = 77, discordance rate = 1.30%
P/LP
B/LB
0
25
9
1
B/LB
P/LP
ClinGen classification

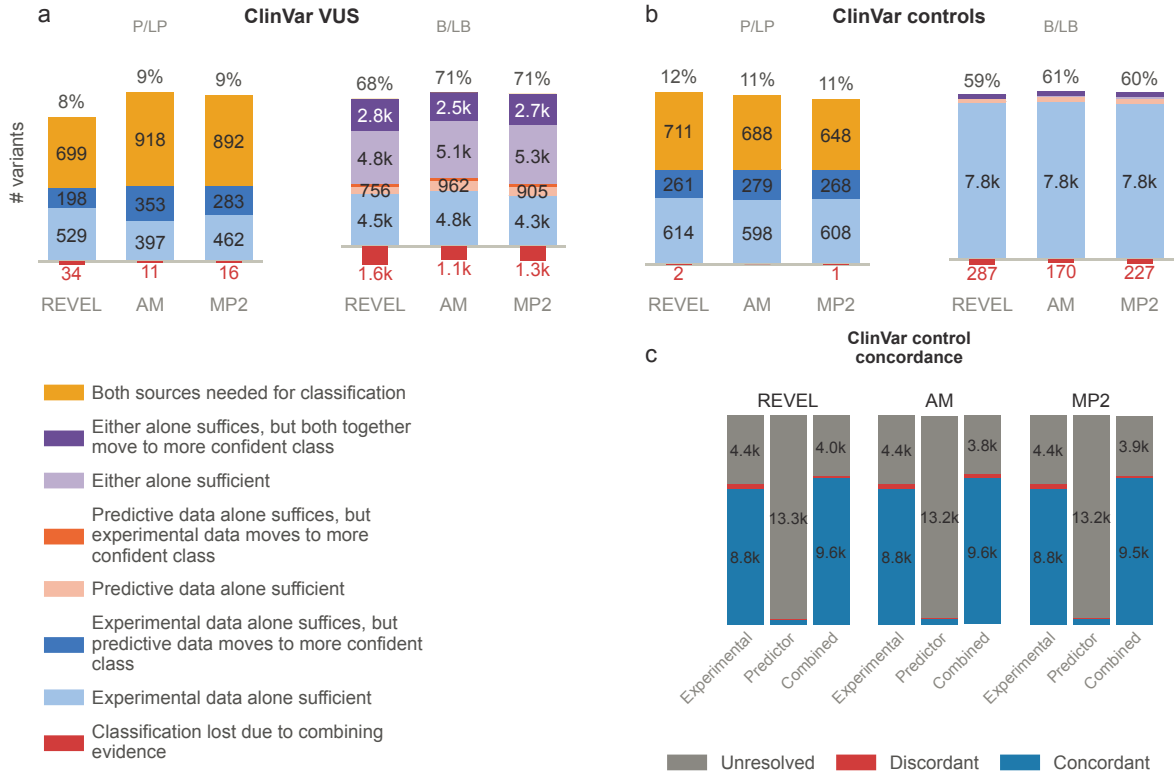

ClinVar controls
All variants
Missense only
a REVEL
b AlphaMissense
c MutPred2
d REVEL
e AlphaMissense
f MutPred2
P
LP
VUS
LB
B
Total = 12,814, discordance rate = 0.75%
Total = 12,848, discordance rate = 1.70%
Total = 11,480, discordance rate = 0.78%
Total = 2,592, discordance rate = 2.16%
Total = 2,626, discordance rate = 6.82%
Total = 1,262, discordance rate = 3.88%
Exp + Pred classification
P/LP
B/LB
3
1,470
7,493
93
5
1,461
7,550
214
3
853
7,481
86
1
1,026
359
55
3
1,021
416
176
1
409
347
48
B/LB P/LP
ClinVar classification
ClinGen Evidence Repository
All variants
Missense only
g REVEL
h AlphaMissense
i MutPred2
j REVEL
k AlphaMissense
l MutPred2
Total = 388, discordance rate = 2.06%
Total = 395, discordance rate = 11.14%
Total = 121, discordance rate = 2.48%
Total = 344, discordance rate = 1.74%
Total = 351, discordance rate = 11.97%
Total = 77, discordance rate = 1.30%
0
153
15
8
0
123
22
44
0
34
14
3
0
144
10
6
0
114
17
42
0
25
9
1
ClinGen classification

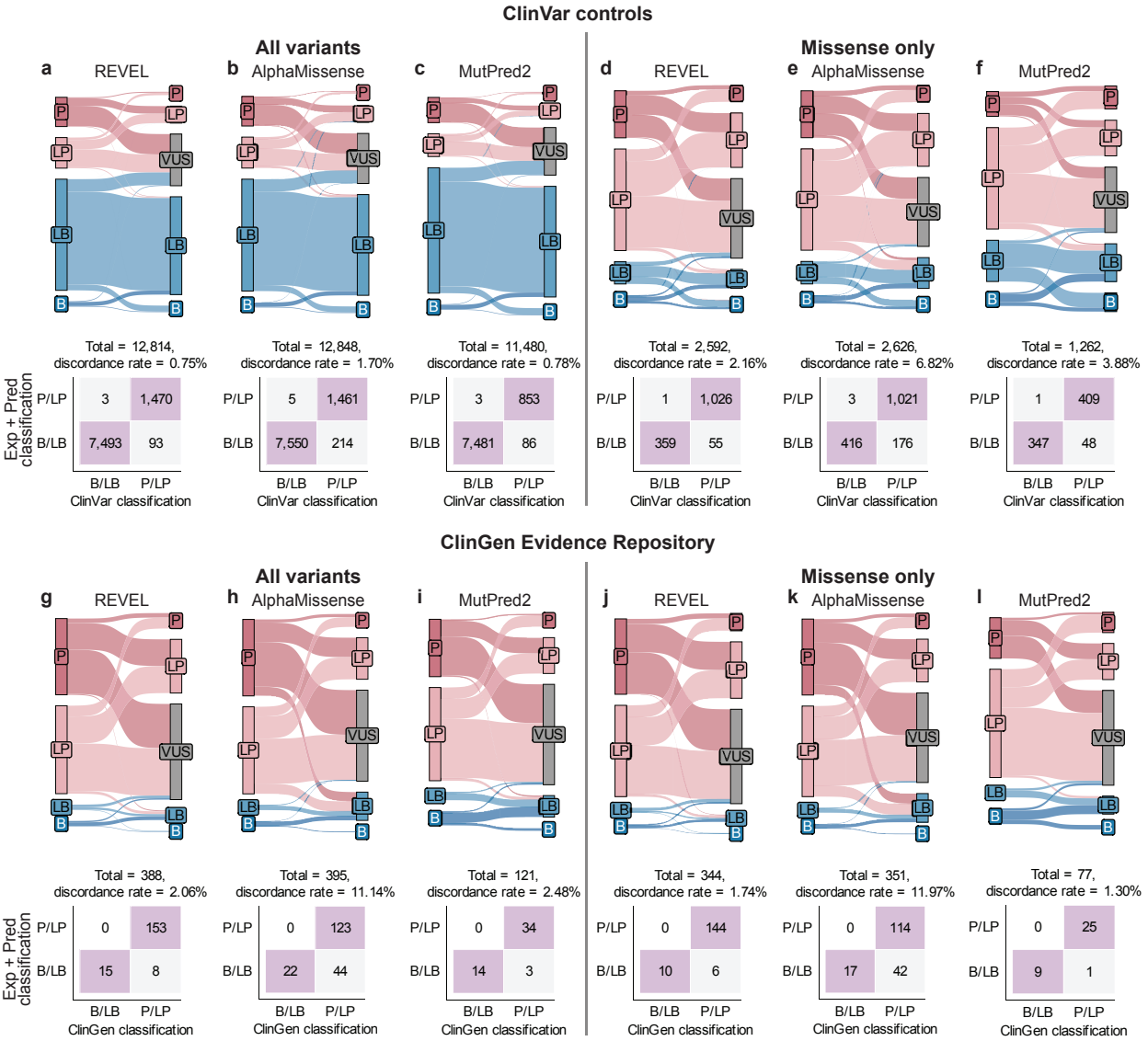
a
REVEL
Expected discordance (LOO)
TP53
DDX3X
F9
VHL
LDLR
Discordance rate
Variants per gene
-log10(p-value)
b
AlphaMissense
Expected discordance (LOO)
TARDBP
LDLR
TP53
F9
G6PD
DDX3X
Discordance rate
Variants per gene
-log10(p-value)
c
MutPred2
Expected discordance (LOO)
TP53
VHL
LDLR
DDX3X
Discordance rate
Variants per gene
-log10(p-value)


ClinVar controls
All variants
Missense only
a REVEL
b AlphaMissense
c MutPred2
d REVEL
e AlphaMissense
f MutPred2
P
LP
VUS
LB
B
Total = 10,752, discordance rate = 1.06%
Total = 10,779, discordance rate = 1.34%
Total = 9,720, discordance rate = 0.87%
Total = 2,089, discordance rate = 3.59%
Total = 2,116, discordance rate = 4.96%
Total = 1,061, discordance rate = 4.34%
Exp + Pred classification
P/LP
B/LB
2 782
6,563 112
1 746
6,604 143
0 307
6,540 85
2 781
338 73
1 746
379 104
0 300
315 46
B/LB P/LP
ClinVar classification
ClinGen Evidence Repository
All variants
Missense only
g REVEL
h AlphaMissense
i MutPred2
j REVEL
k AlphaMissense
l MutPred2
Total = 285, discordance rate = 3.16%
Total = 291, discordance rate = 5.84%
Total = 98, discordance rate = 8.16%
Total = 255, discordance rate = 3.14%
Total = 261, discordance rate = 6.13%
Total = 68, discordance rate = 10.29%
1 118
12 8
0 79
15 17
0 21
9 8
1 118
11 7
0 79
14 16
0 21
8 7
ClinGen classification



Extended Data Figure 7: Ablation analysis


a
ClinVar VUS
P/LP
B/LB
variants
8%
9%
9%
699
198
529
34
918
353
397
11
892
283
462
16
REVEL
AM
MP2
68%
71%
71%
2.8k
4.8k
756
4.5k
1.6k
2.5k
5.1k
962
4.8k
1.1k
2.7k
5.3k
905
4.3k
1.3k
b
ClinVar controls
12%
11%
11%
711
261
614
2
688
279
598
648
268
608
1
59%
61%
60%
7.8k
287
170
227
c
ClinVar control concordance
REVEL
AM
MP2
4.4k
8.8k
13.3k
4.0k
9.6k
4.4k
8.8k
13.2k
3.8k
9.6k
4.4k
8.8k
13.2k
3.9k
9.5k
Experimental
Predictor
Combined
Both sources needed for classification
Either alone suffices, but both together move to more confident class
Either alone sufficient
Predictive data alone suffices, but experimental data moves to more confident class
Predictive data alone sufficient
Experimental data alone suffices, but predictive data moves to more confident class
Experimental data alone sufficient
Classification lost due to combining evidence
Unresolved
Discordant
Concordant

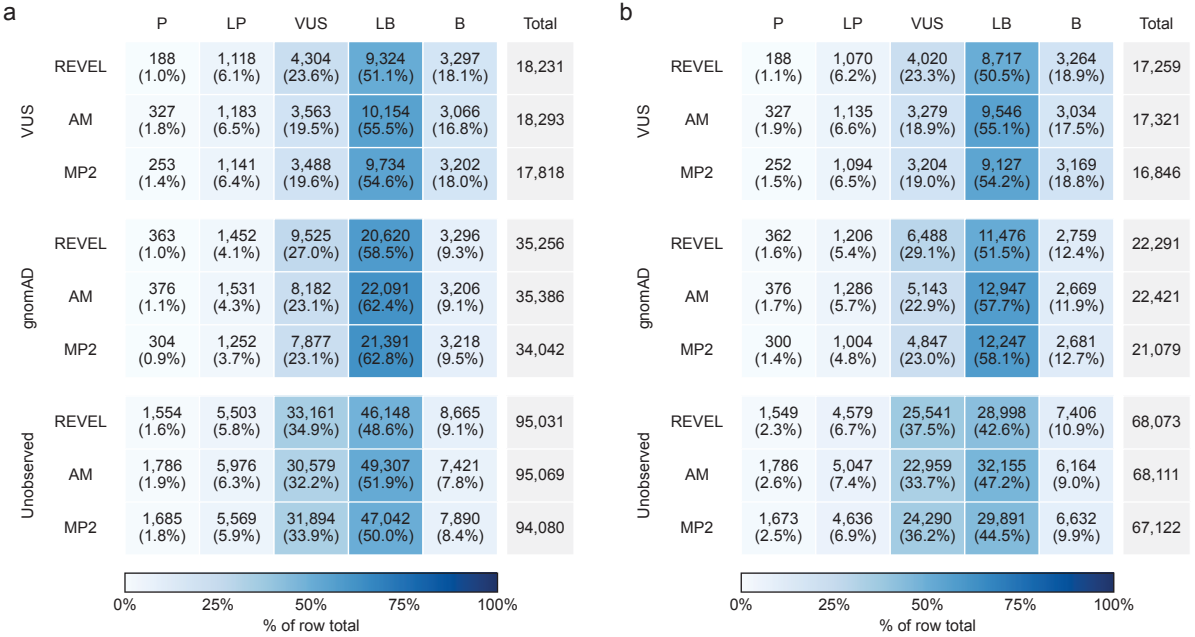


Extended Data Figure 3: Classifications of ClinVar and Clingen Evidence Repository

Experimental evidence calibrated with ExCALIBR (except for F9 and TP53 where a model was applied) and predictive evidence calibrated with the gene-specific approach when available or the genome-wide approach for the remaining genes applied to control variants from ClinVar and the ClinGen Evidence Repository.

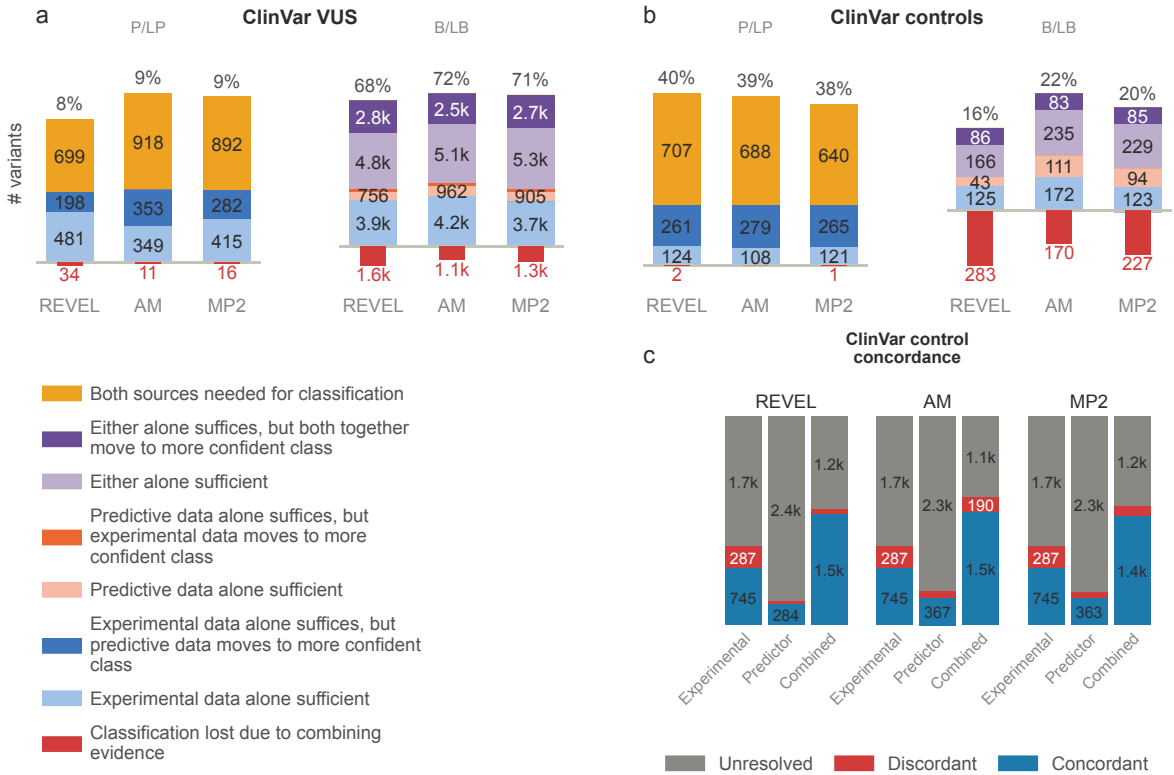
Sankey diagrams showing the relationship of ClinVar variant classifications and their classifications produced by combining only experimental evidence and evidence from REVEL, AlphaMissense, and MutPred2 for all variants (a-c) and missense variants only (d-f). Paired confusion matrices comparing original ClinVar classifications to classifications assigned using experimental and predictive evidence only summarizing concordant (lilac) (i.e., agreeing with original BLB and PLP ClinVar classifications) and discordant (grey) variant classifications (lipped from BLB to PLP from the original ClinVar classification or vice versa) for the indicated predictor.

Sankey diagrams showing the relationship of control variants from the ClinGen Evidence Repository (v2.5.6), using updated classifications in which experimental and predictive evidence were removed, to classifications assigned using experimental and predictive evidence only, using REVEL, AlphaMissense, and MutPred2 with paired confusion matrices for all variants (g-i) and missense only (j-l).



Extended Data Figure 6: Calibrated experimental and predictive data provide sufficient evidence to resolve VUS and preclassify gnomAD variants and previously unobserved variants.

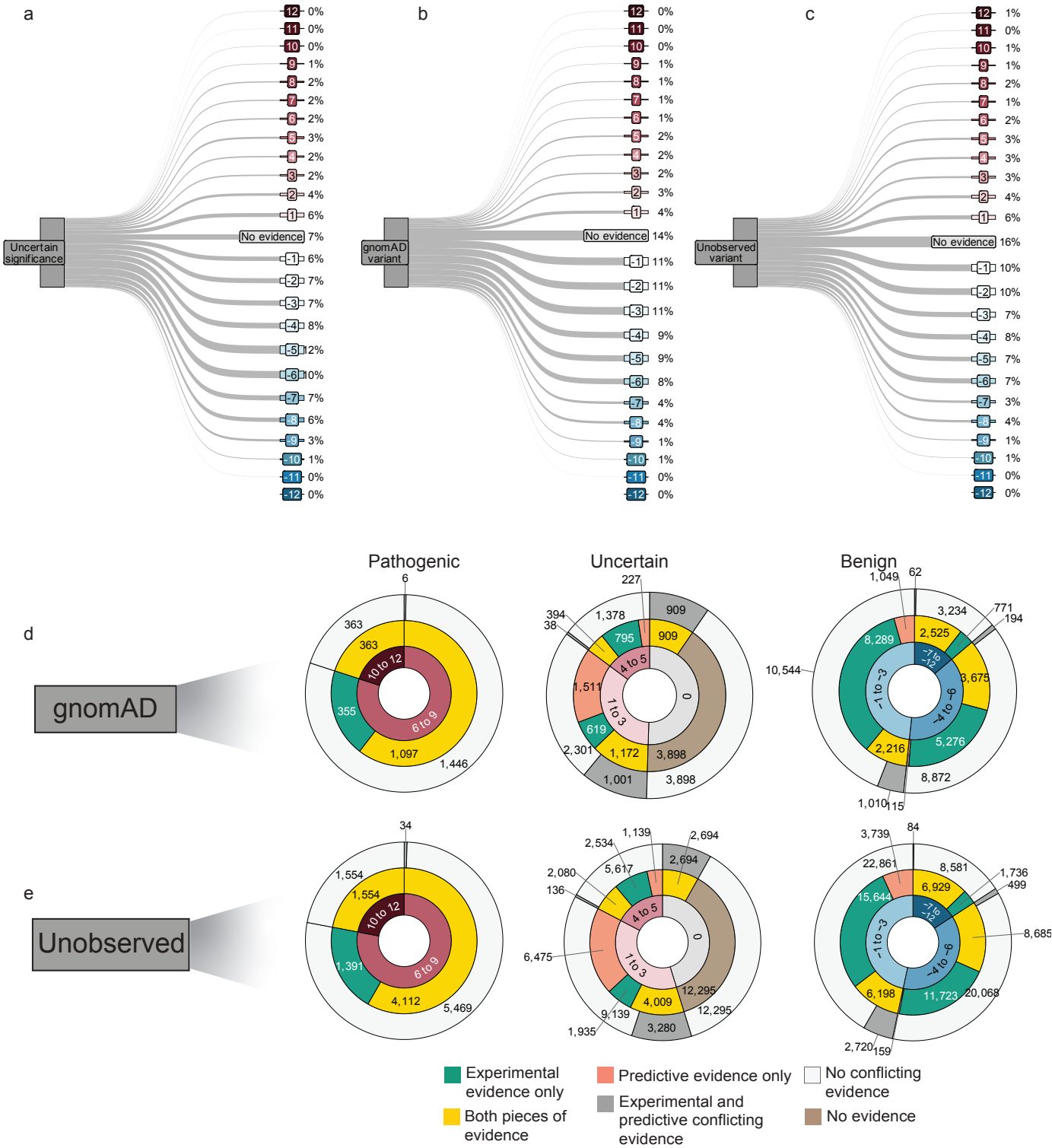
Heatmaps showing the fraction of VUS (ClinVar) or SNVs (gnomAD, unobserved) that can be classified to pathogenic (P), likely pathogenic (LP), likely benign (LB), benign (B) or VUS using our scalable workflow combining calibrated experimental and predictive evidence from the indicated predictor across different variant sources ClinVar VUS (top), gnomAD (middle) or previously unobserved variants (bottom) for (a) all variants and (b) missense only.



Extended Data Figure 7: Ablation analysis

An ablation analysis to assess the contribution of experimental and predictive evidence to classification of ClinVar missense VUS (a) and missense control variants (b). Stacked bar plots show the number of variants for which experimental or predictive sufficed alone, both were required for classification (neither alone), either sufficed but combined evidence moved the classification to a more confident class (LB -> B or LP -> P), or the combination withdrew a classification produced by one source, as indicated for PLP classifications (left) and BLB classifications (right) stratified by predictor (AM = AlphaMissense, MP2 = MutPred2). The total percentage of variants that reach a definitive PLP or BLB class is given on top of the barplot.

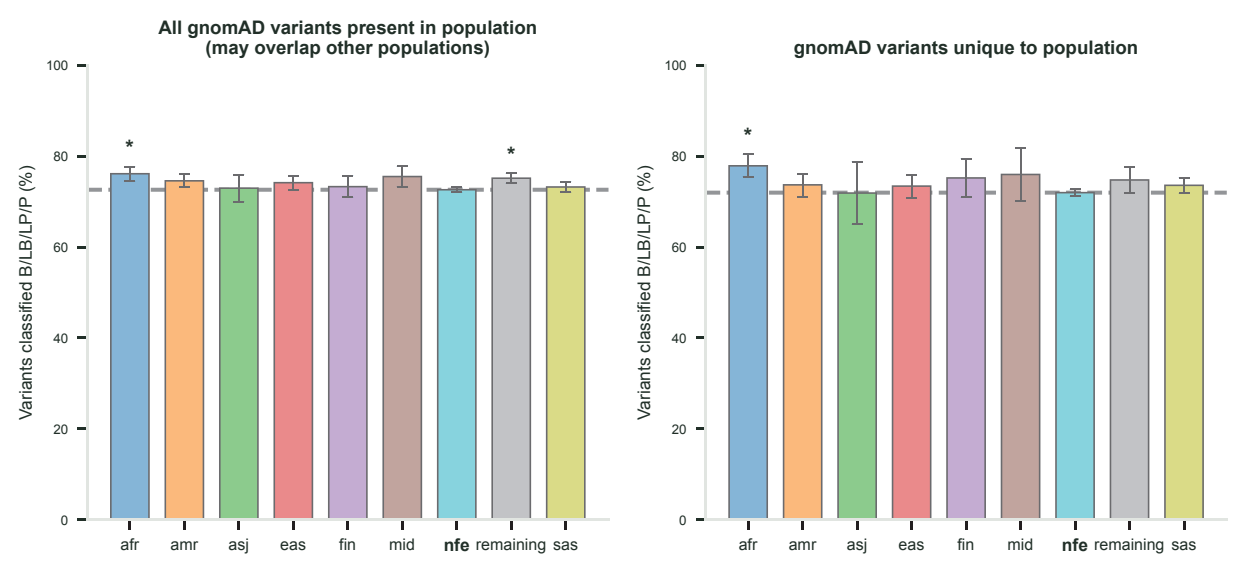
A control-concordance analysis restricted to ClinVar missense control variants (c), which tests whether classifications based on experimental and predictive evidence or both agree with classifications from ClinVar. For each predictor (REVEL, AM = AlphaMissense, MP2 = MutPred2), three bars show the outcome when only functional evidence, only predictor evidence, or both combined are used to classify variants; each bar is divided into the proportion of variants classified concordantly (resolved in agreement with the known ClinVar direction), classified discordantly (resolved in the opposite direction), or unresolved (evidence did not cross either classification threshold).

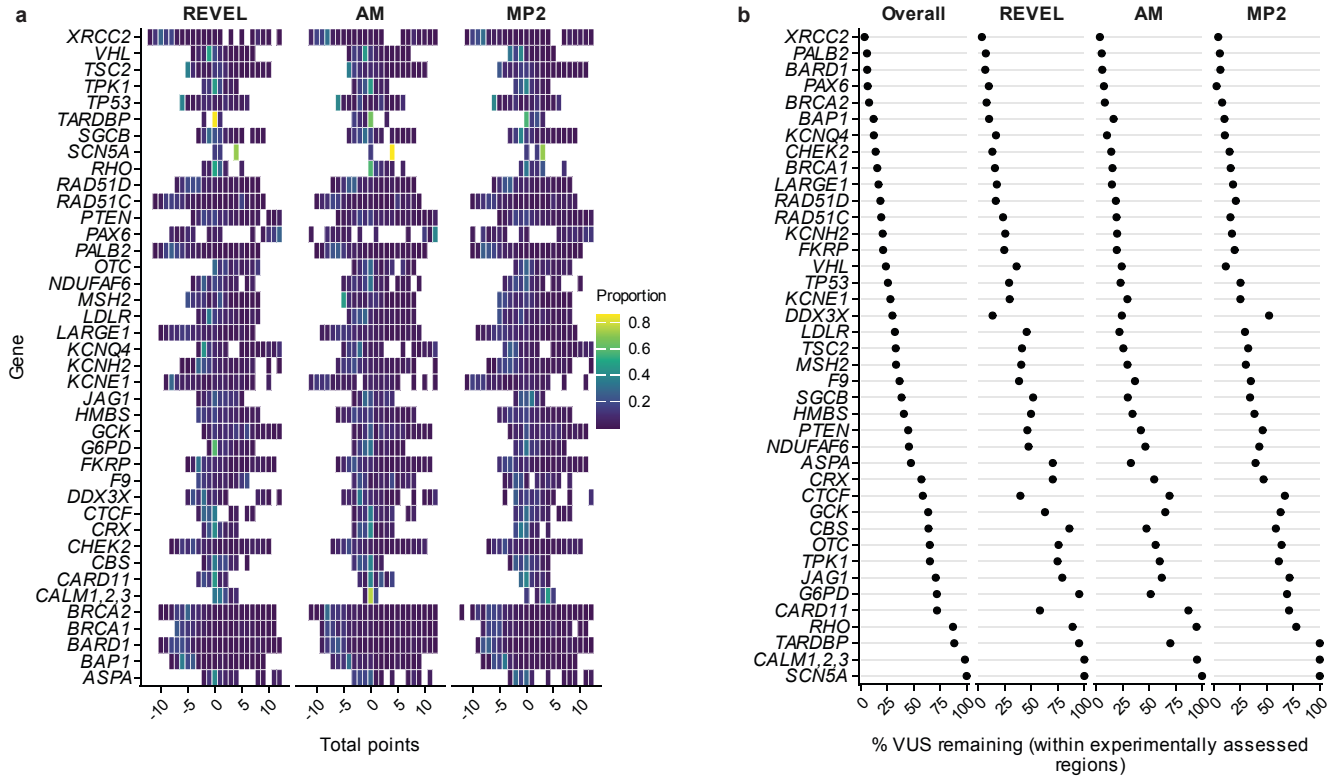


Extended Data Figure 8: Distribution of variant evidence points across ClinVar VUS, gnomAD and unobserved variants

(a–c) Sankey diagrams illustrate the distribution of variants across total evidence point categories (experimental and predictive evidence (REVEL)) for (a) 18,231 ClinVar variants of uncertain significance (VUS), (b) 35,256 gnomAD variants, and (c) 95,031 unobserved variants. The width of each flow is proportional to the number of variants assigned to the corresponding evidence point category, and the percentage of variants at each point value is shown.

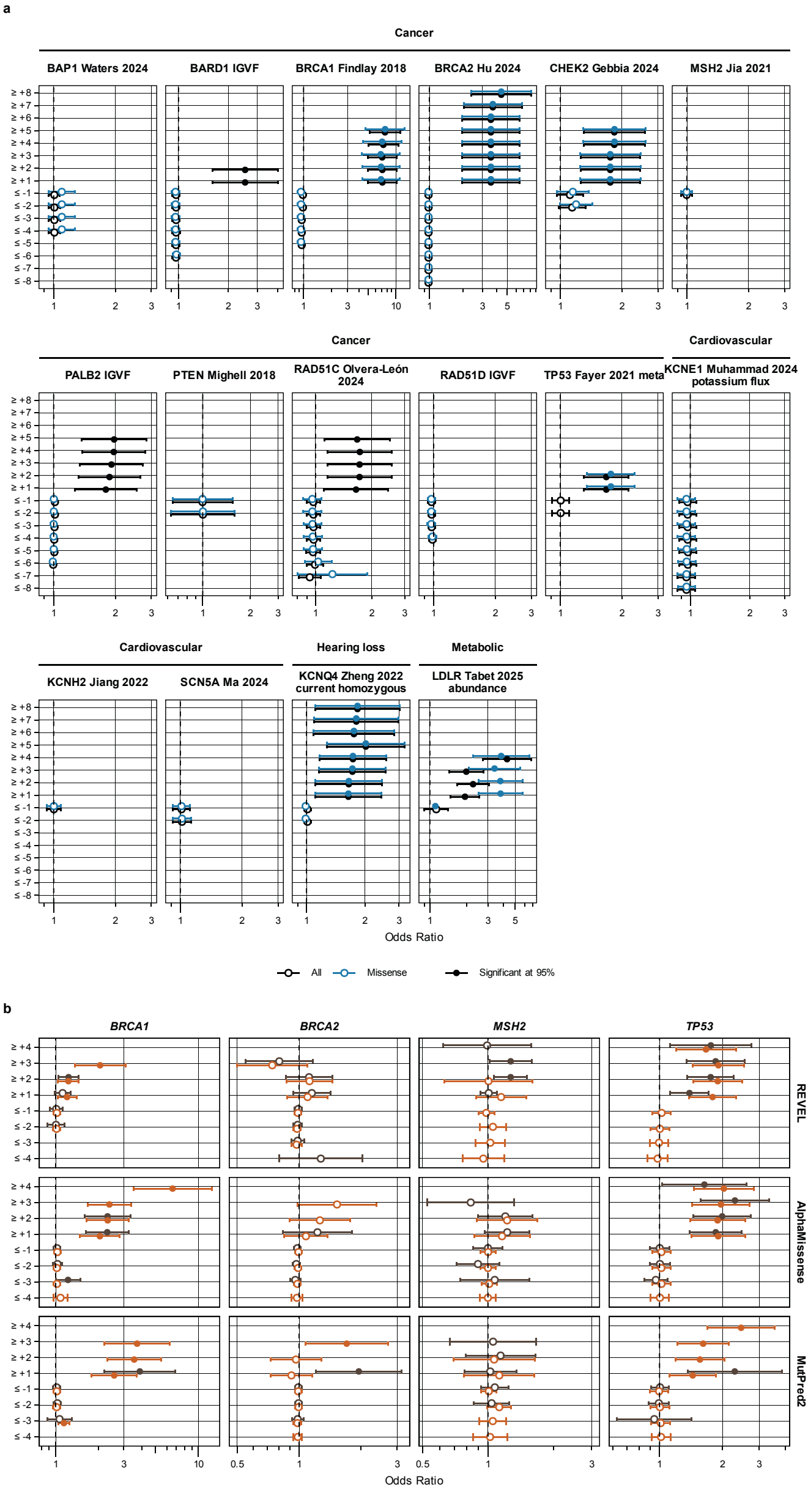
(d,e) Multi-ring donut plots summarizing the evidence source and points for (d) gnomAD variants and (e) unobserved variants. Variants are grouped into pathogenic, uncertain, and benign categories. The innermost ring represents total evidence point ranges, the middle ring denotes the source of evidence (experimental, predictive, or both), and the outermost ring indicates concordance or conflict between evidence sources.

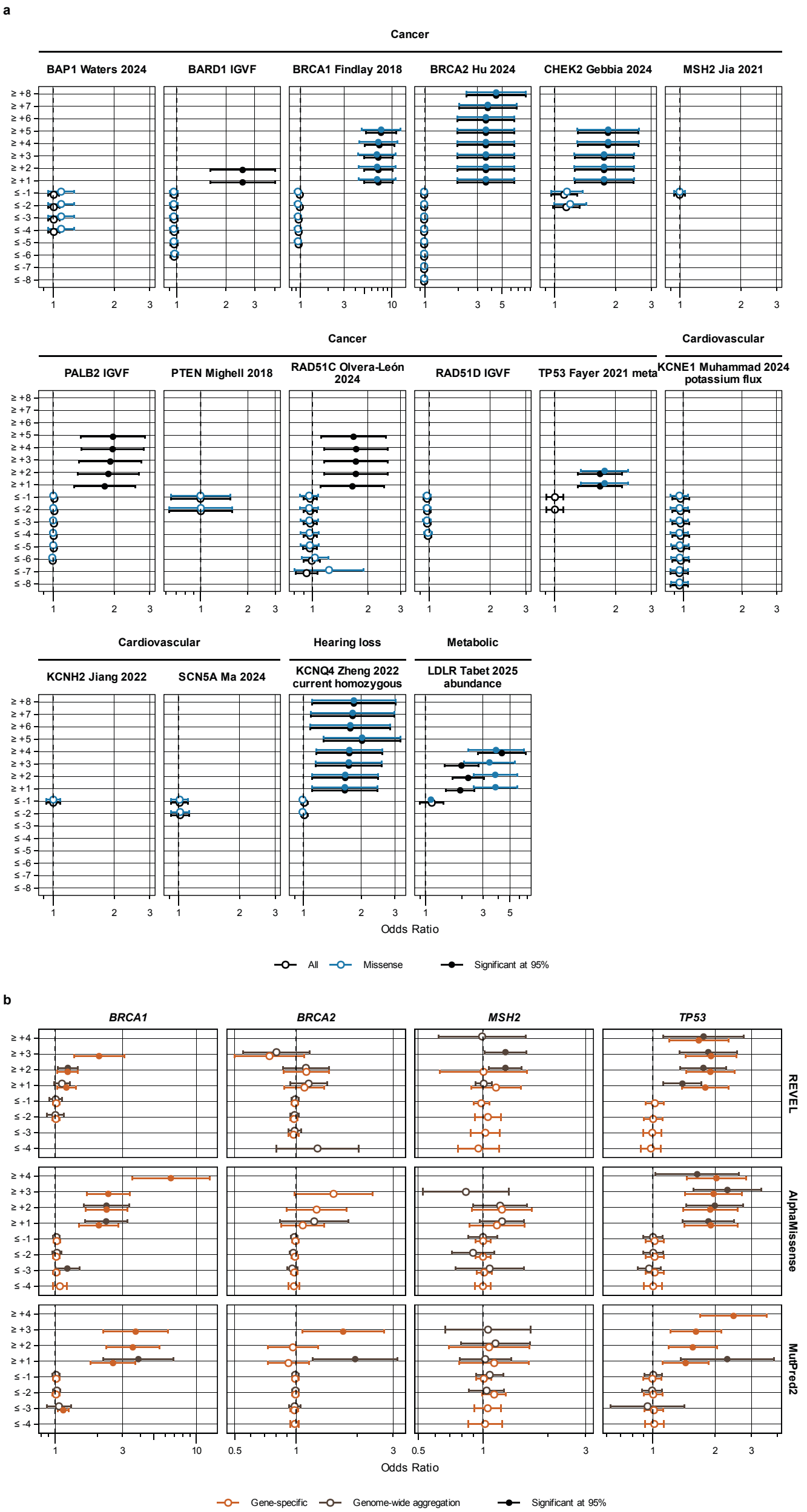

All gnomAD variants present in population
(may overlap other populations)
Variants classified B/LB/LP/P (%)
100
80
60
40
20
0
*
*
afr
amr
asj
eas
fin
mid
nfe
remaining
sas
gnomAD variants unique to population
Variants classified B/LB/LP/P (%)
100
80
60
40
20
0
*
afr
amr
asj
eas
fin
mid
nfe
remaining
sas



Extended Data Figure 9: Predictor-specific distribution of total points and remaining VUS burden across genes

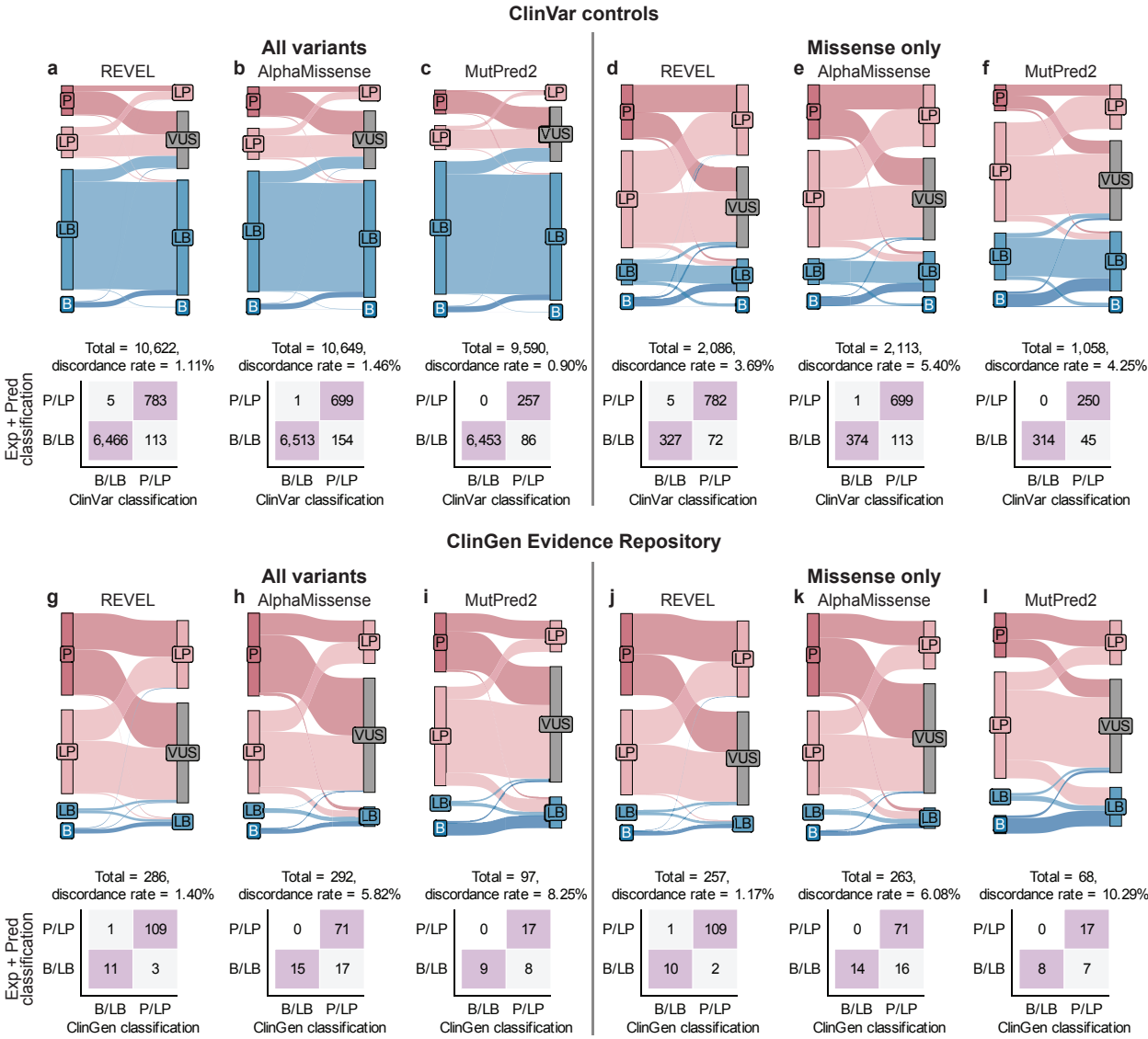
(a) Heatmaps showing the distribution of total combined evidence points (experimental and predictive evidence) across variants for each gene, stratified by predictor (REVEL, AlphaMissense, MutPred2). For each gene, tile color denotes the proportion of variants assigned to a given total point value, normalized within the gene. Darker shading (towards blue) indicates a lower fraction of variants occupying that point range, highlighting gene-and predictor-specific differences in how variants are distributed across total combined points (b) Rank-based dot plots showing the percentage of variants remaining uncertain (0–5 total points) for each gene (within experimentally assayed regions). Genes are ordered by the overall VUS remaining (computed across predictors), and panels display the corresponding value for each individual predictor (REVEL, AlphaMissense, MutPred2).


a
Cancer
BAP1 Waters 2024
BARD1 IGVF
BRCA1 Findlay 2018
BRCA2 Hu 2024
CHEK2 Gebbia 2024
MSH2 Jia 2021
Cancer
Cardiovascular
PALB2 IGVF
PTEN Mighell 2018
RAD51C Olvera-León 2024
RAD51D IGVF
TP53 Fayer 2021 meta
KCNE1 Muhammad 2024 potassium flux
Cardiovascular
Hearing loss
Metabolic
KCNH2 Jiang 2022
SCN5A Ma 2024
KCNQ4 Zheng 2022 current homozygous
LDLR Tabet 2025 abundance
≥ +8
≥ +7
≥ +6
≥ +5
≥ +4
≥ +3
≥ +2
≥ +1
≤ -1
≤ -2
≤ -3
≤ -4
≤ -5
≤ -6
≤ -7
≤ -8
Odds Ratio
All
Missense
Significant at 95%
b
BRCA1
BRCA2
MSH2
TP53
REVEL
AlphaMissense
MutPred2
≥ +4
≥ +3
≥ +2
≥ +1
≤ -1
≤ -2
≤ -3
≤ -4
Odds Ratio
Gene-specific
Genome-wide aggregation
Significant at 95%



Extended Data Figure 2: Association of calibrated evidence with disease in the All of Us biobank

Odds ratios (ORs) for disease among individuals carrying variants that meet each evidence strength threshold, shown for gene-disease pairs with sufficient case cohorts (**Supplementary Data 2**). The x axis shows the OR, with a dashed vertical line at OR = 1. The y axis shows variant groups by assigned evidence points. Each group includes all variants reaching at least that number of points, so cohorts shrink toward the extremes. ORs are not shown where the number of carrier cases or carrier controls fell below 21 individuals. Circles show estimated ORs and whiskers show 95% confidence intervals. Filled circles indicate ORs significant at the 95% level by a two-sided Wald test. (a) ORs for variants grouped by ExCALIBR-calibrated experimental evidence from one representative functional study per gene (**Supplementary Data 2**), except for TP53, where points were assigned using the OddsPath calibration from the meta-analysis of Fayer et al. 2021. ORs are shown for all variants (black) and missense variants only (blue). (b) ORs for variants grouped by calibrated computational predictor evidence (REVEL, AlphaMissense and MutPred2) for BRCA1, BRCA2, MSH2 and TP53, comparing gene-specific calibration (orange) with genome-wide aggregate calibration (brown).

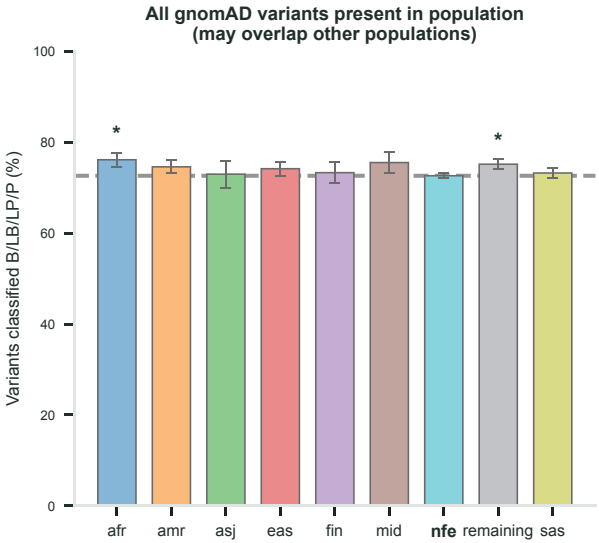


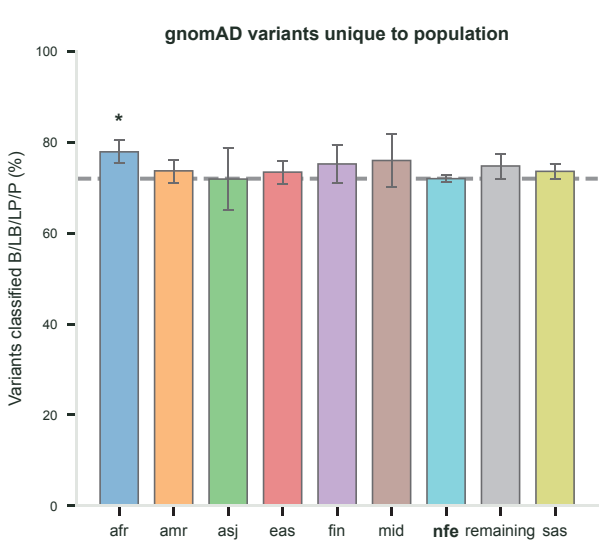
Extended Data Figure 5: Classifications of control variants from ClinVar or the ClinGen Evidence Repository using only ClinGen-accepted calibration methods

Experimental evidence calibrated using OddsPath and predictive evidence calibrated using the genome-wide approach applied to control variants from ClinVar and the ClinGen Evidence Repository. Genes are subset to the 31 with thresholds for functionally normal vs. abnormal variants provided by authors (Supplementary Data 3, column: 'Functional Classification Provided?').

Sankey diagrams showing the relationship of ClinVar variant classifications and their classifications produced by combining only experimental evidence from OddsPath calibrations and genome-wide predictive evidence from REVEL, AlphaMissense, and MutPred2 for all variants (a-c) and missense variants only (d-f). Paired confusion matrices comparing original ClinVar classifications to classifications assigned using experimental and predictive evidence only (OddsPath and genome-wide predictive evidence) summarizing concordant (lilac) (i.e., agreeing with original BLB and PLP ClinVar classifications) and discordant (grey) (flipped from BLB to PLP from the original ClinVar classification or vice versa) variant classifications for the indicated predictor.

Sankey diagrams showing the relationship of control variants from the ClinGen Evidence Repository (v2.5.6), using updated classifications in which experimental and predictive evidence were removed, to classifications assigned using experimental and predictive evidence only, using REVEL, AlphaMissense, and MutPred2 with paired confusion matrices for all variants (g-i) and missense only (j-l).

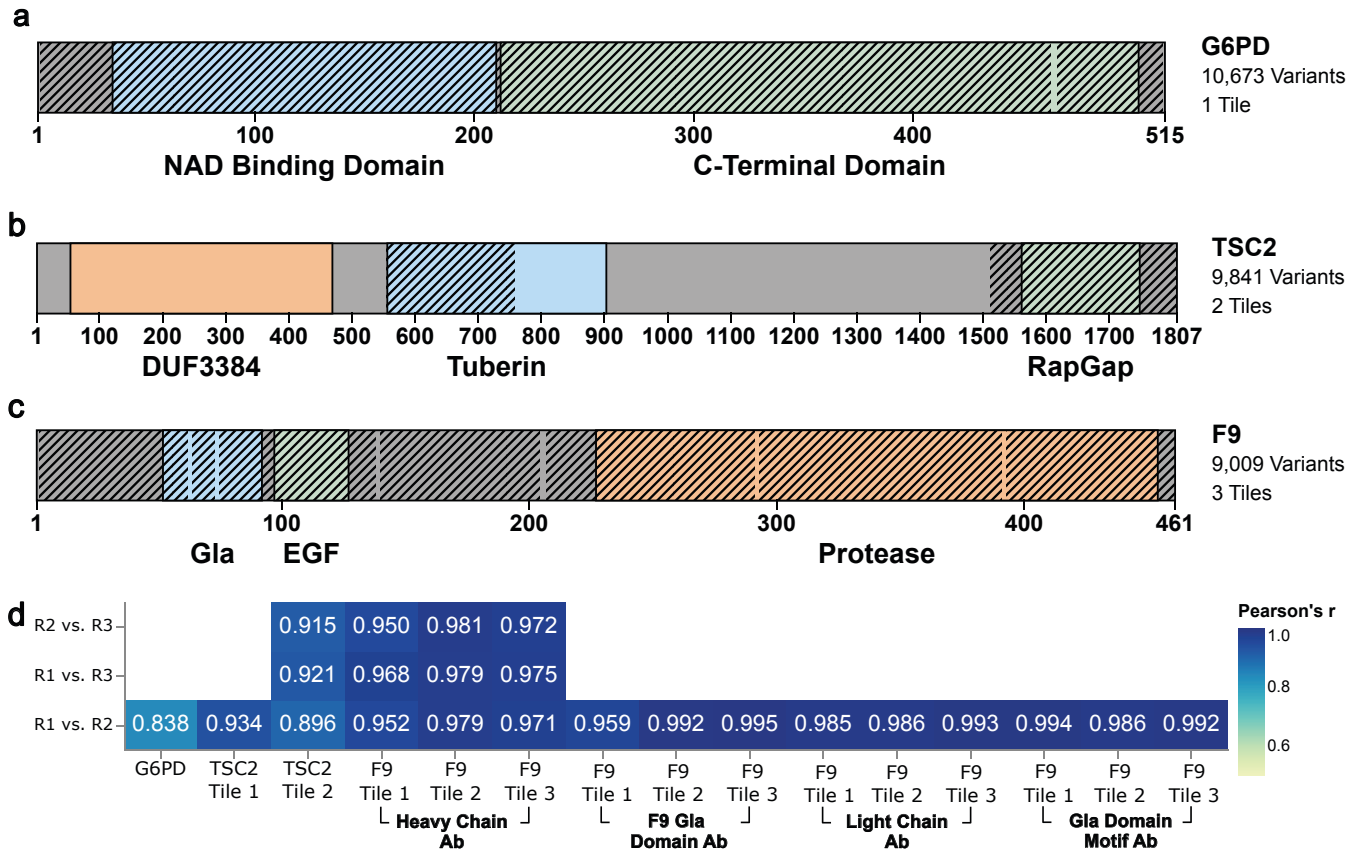


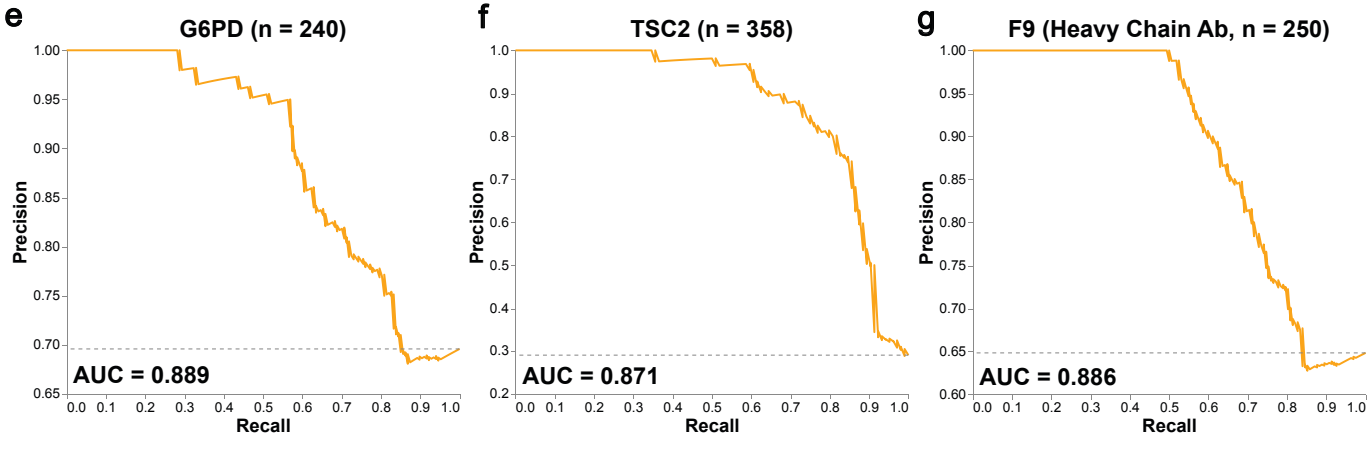


Extended Data Figure 10: Classification rates are similar across gnomAD genetic ancestry groups.

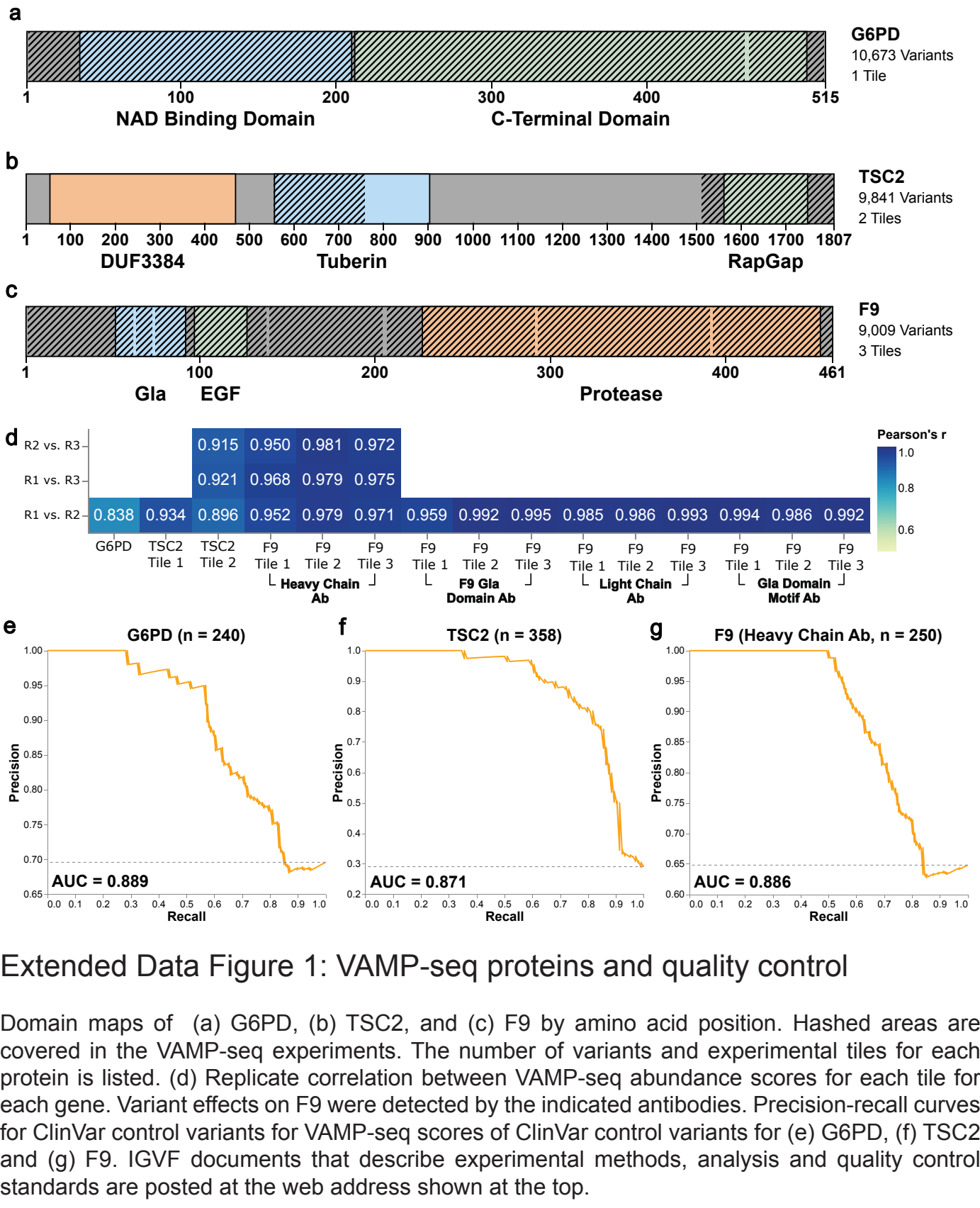
Percentage of gnomAD variants reaching a determinate classification (benign, likely benign, likely pathogenic or pathogenic) using only experimental and predictive evidence, shown for all variants observed in each genetic ancestry group (left; a variant may occur in more than one group) and for variants observed only in one group (right). n indicates the number of variants per group. Ancestry-specific frequencies were available for 26,687 of 35,256 gnomAD variants. Error bars show 95% bootstrap confidence intervals (1,000 resamples). The dashed line marks the classification rate for the non-Finnish European (NFE) group. Asterisks indicate rates significantly different from NFE (Bonferroni-corrected $p < 0.05$). No group had a statistically significantly lower classification rate than NFE. Classification rates were modestly higher for African/African American genetic ancestry (AFR) among all variants (76.2% vs 72.9%, corrected $p = 0.00066$) and among unique variants (78.1% vs 72.4%, corrected $p = 0.00058$), and for the remaining-individuals group among all variants (75.2% vs 72.9%, corrected $p = 0.0089$). Remaining individuals are defined as those that cannot be confidently assigned to a primary global ancestry group.

IGVF Standards Documents for VAMP-seq & MultiSTEP:
https://data.igvf.org/methodology/experimental_standards/vamp-seq-standards/



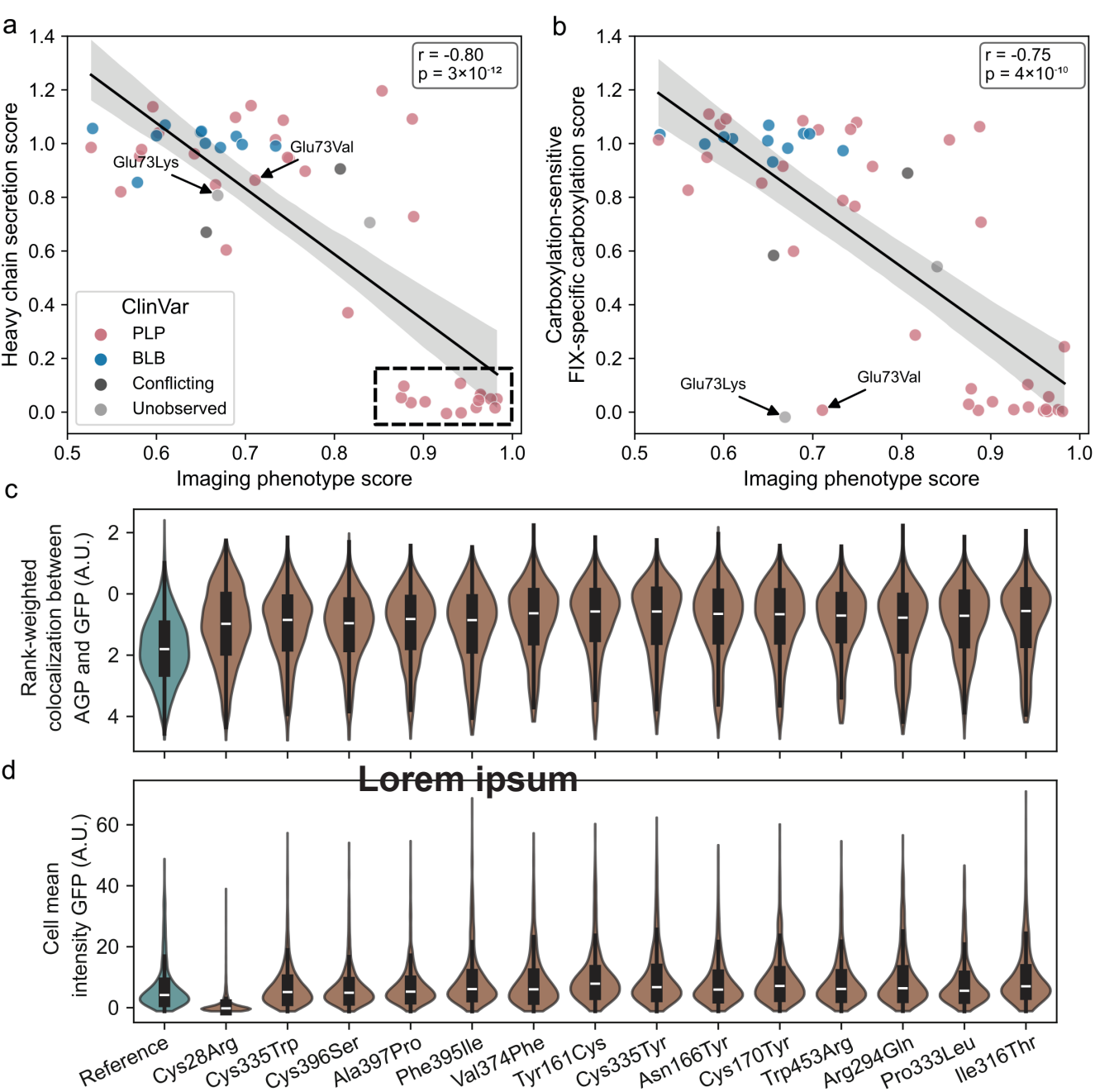


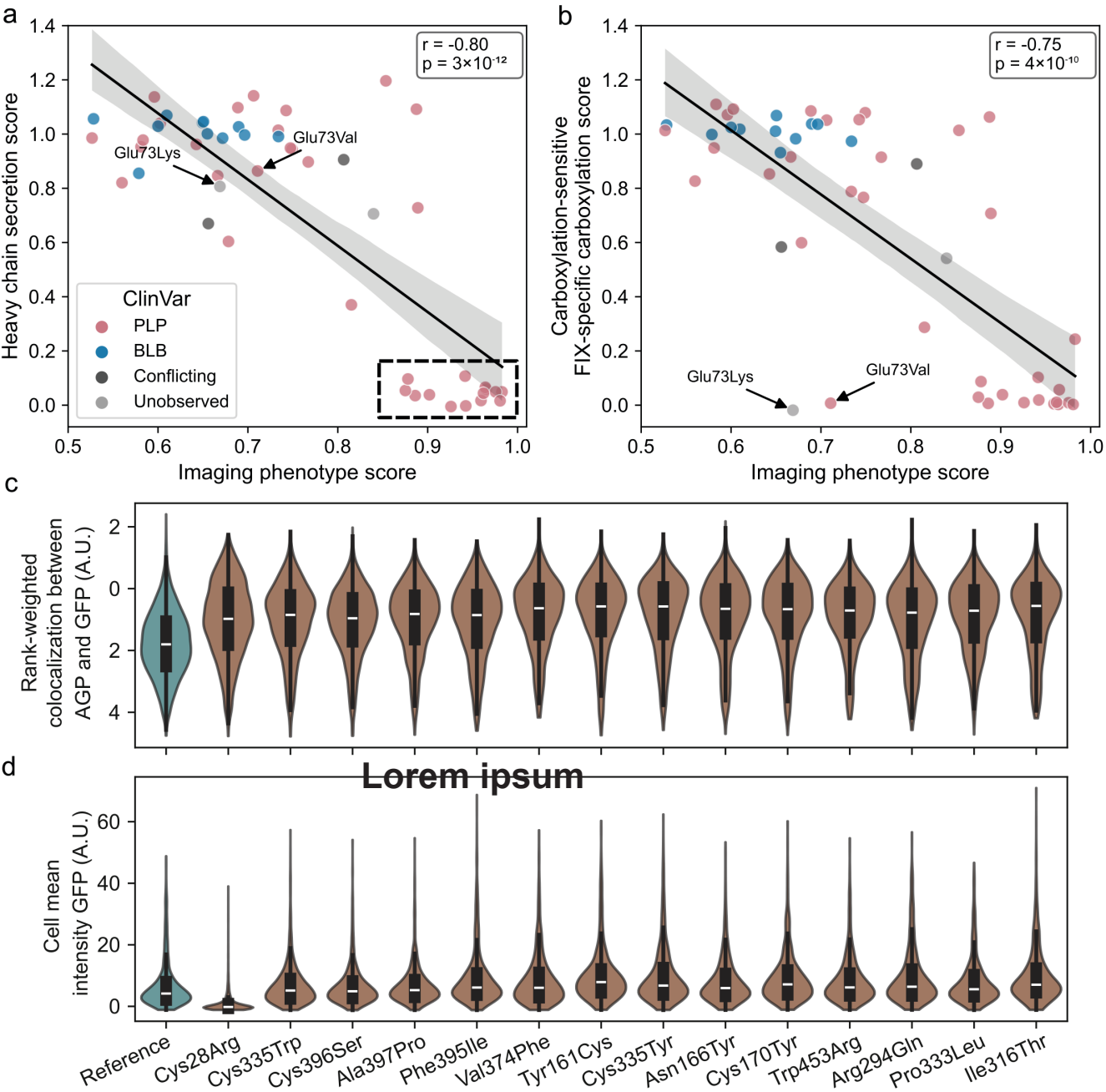
IGVF Standards Documents for VAMP-seq & MultiSTEP:
https://data.igvf.org/methodology/experimental_standards/vamp-seq-standards/



Extended Data Figure 1: VAMP-seq proteins and quality control

Domain maps of (a) G6PD, (b) TSC2, and (c) F9 by amino acid position. Hashed areas are covered in the VAMP-seq experiments. The number of variants and experimental tiles for each protein is listed. (d) Replicate correlation between VAMP-seq abundance scores for each tile for each gene. Variant effects on F9 were detected by the indicated antibodies. Precision-recall curves for ClinVar control variants for VAMP-seq scores of ClinVar control variants for (e) G6PD, (f) TSC2 and (g) F9. IGVF documents that describe experimental methods, analysis and quality control standards are posted at the web address shown at the top.


a
r = -0.80
p = 3×10⁻¹²
Glu73Lys
Glu73Val
ClinVar
PLP
BLB
Conflicting
Unobserved
Heavy chain secretion score
Imaging phenotype score
b
r = -0.75
p = 4×10⁻¹⁰
Glu73Lys
Glu73Val
Carboxylation-sensitive
FIX-specific carboxylation score
Imaging phenotype score
c
Rank-weighted
colocalization between
AGP and GFP (A.U.)
d
Lorem ipsum
Cell mean
intensity GFP (A.U.)
Reference
Cys28Arg
Cys335Trp
Cys396Ser
Ala397Pro
Phe395Ile
Val374Phe
Tyr161Cys
Cys335Tyr
Asn166Tyr
Cys170Tyr
Trp453Arg
Arg294Gln
Pro333Leu
Ile316Thr



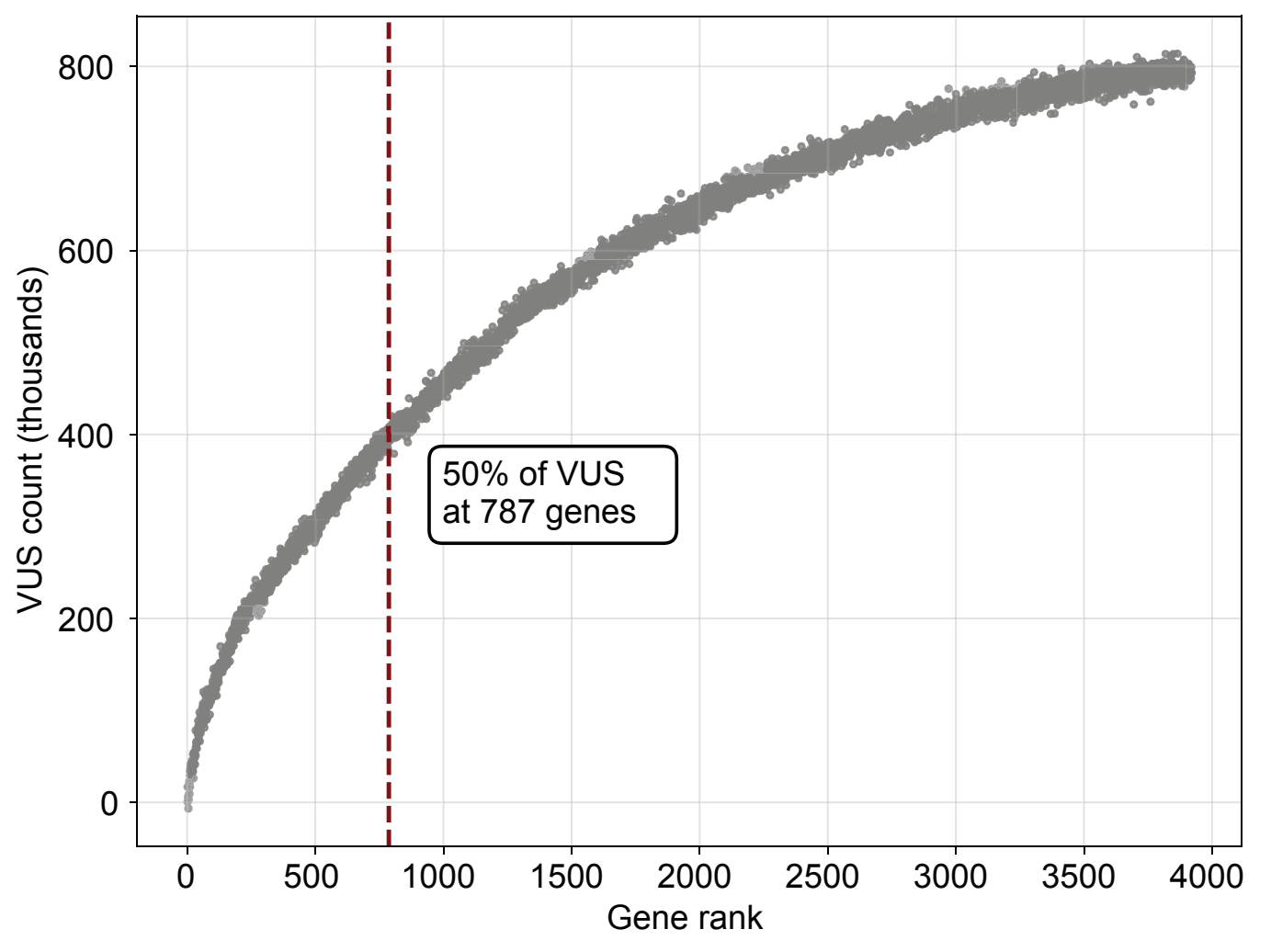
Extended Data Figure 11: Variant painting reveals mechanism for low abundance variants identified by VAMP-seq

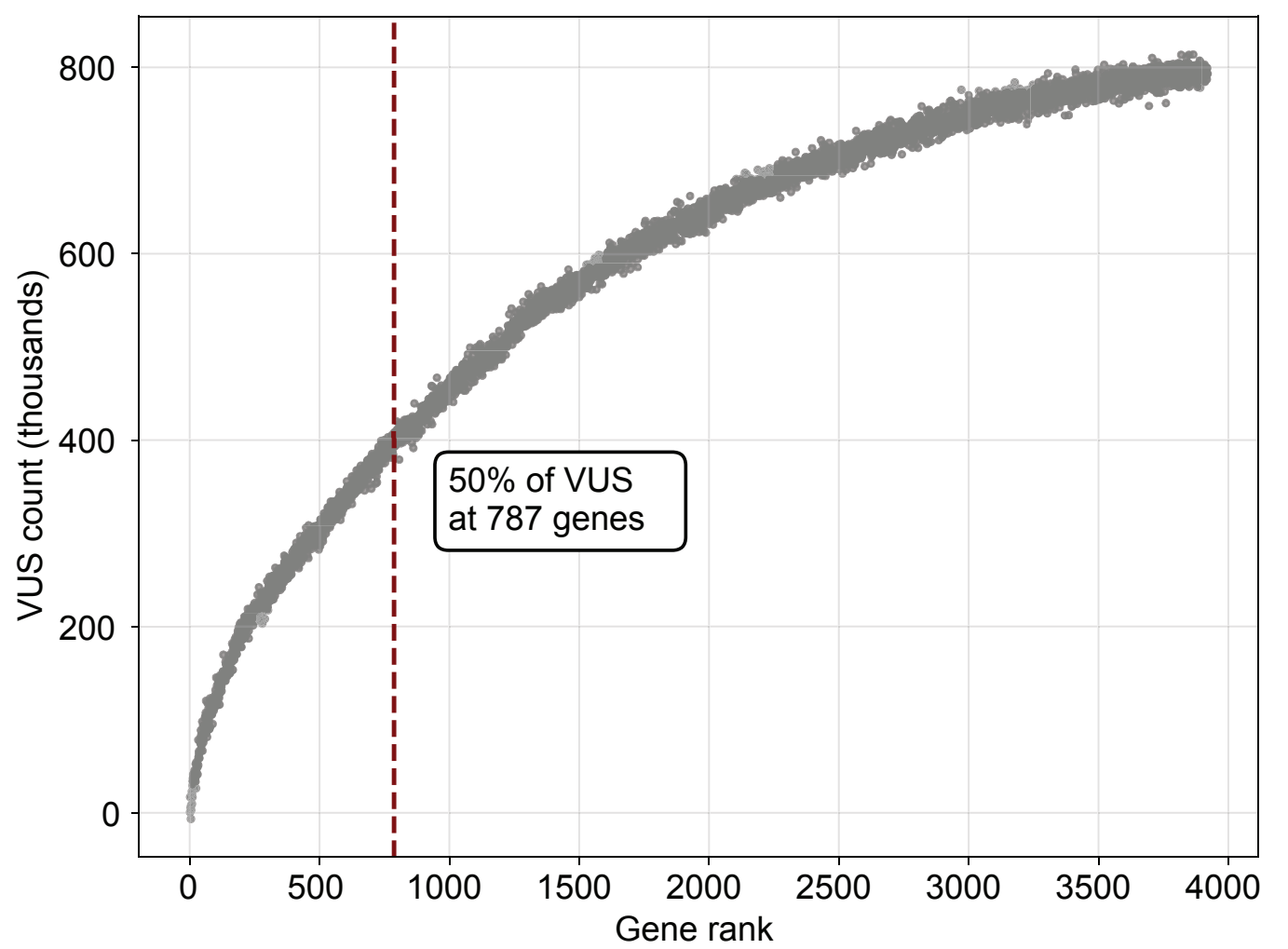
(a) Correlation between imaging phenotype scores and factor IX heavy-chain secretion scores. Boxed variants (n=14) are poorly secreted and show increased Golgi co-localization, indicating secretory pathway retention.

(b) Correlation between imaging phenotype scores and factor IX-specific carboxylation score.

(c) Distribution of single-cell normalized values for rank-weighted colocalization between AGP staining (actin, Golgi, plasma membrane markers) and GFP-tagged variants with VAMP-seq abundance score < 0.2.

(d) Distribution of single-cell normalized values for mean GFP intensity per cell, across the factor IX reference allele and selected variants VAMP-seq abundance score < 0.2.


800
600
400
200
0
VUS count (thousands)
50% of VUS
at 787 genes
0
500
1000
1500
2000
2500
3000
3500
4000
Gene rank



Extended Data Figure 12: The number of disease-associated genes needed to reclassify 50% of VUS

Collector's curve for the number of VUS (single nucleotide variants) ranked by gene (ClinVar, January 2025) for the ~4,500 genes with at least a moderate level of gene-disease validity according to GenCC (https://thegencc.org/). The vertical dashed line represents the ranked gene count where 50% of VUS are collected.